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1. INTRODUCTION

1.1 Background and Motivation

The film industry is one of the most dynamic and high-risk sectors, characterized by large financial investments and unpredictable returns. Over the years, the scale of movie production has increased significantly, with producers allocating substantial budgets for production, marketing, distribution, and promotions. Despite these investments, the success of a movie at the box office is influenced by numerous uncertain factors such as audience preferences, competition, critical reviews, timing of release, and overall market trends. As a result, predicting whether a movie will be profitable or not remains a complex challenge for producers and stakeholders.

Traditionally, decisions in the film industry have been based on experience, intuition, and past success of actors, directors, or production houses. However, such approaches are often subjective and may not accurately reflect changing audience behavior and market conditions. With the increasing availability of digital data from platforms such as IMDb, social media, and online streaming services, there is a growing opportunity to apply data-driven techniques to analyze historical patterns and make informed predictions about movie performance.

Data mining and machine learning have gained prominence as effective tools for extracting meaningful insights from large datasets. These techniques enable the identification of relationships between various factors such as movie ratings, budget, genre, cast performance, and audience engagement. By leveraging these insights, predictive models can be developed to estimate box office revenue and evaluate the potential success of a movie before its release.

The motivation behind this study arises from the need to minimize financial risk and improve decision-making in the film industry. Producers often face uncertainty regarding return on investment, especially when dealing with high-budget projects. A reliable prediction model can provide valuable guidance by forecasting expected revenue and indicating whether a project is

likely to be profitable. This not only supports better investment decisions but also enhances planning related to budget allocation, marketing strategies, and release timing.

Furthermore, the integration of machine learning techniques such as Random Forest and Support Vector Machine allows for improved accuracy and robustness in prediction. These models are capable of handling complex relationships within the data and adapting to different patterns, making them suitable for real-world applications in the entertainment industry.

So this study is motivated by the growing need for analytical and data-driven approaches in the film industry. By applying data mining techniques to predict movie success, the project aims to provide a practical solution that reduces uncertainty, supports financial planning, and contributes to more informed and strategic decision-making. In the traditional approach, producers often rely on past experiences, industry trends, and the popularity of actors or directors to make investment decisions. However, such decisions are largely intuitive and may not be reliable in a rapidly changing market environment. Audience preferences are constantly evolving, and factors that contributed to success in the past may not guarantee similar results in the future. Therefore, there is a growing need for systematic and analytical methods to evaluate potential outcomes more accurately.

With the advancement of technology and the availability of large datasets, data mining has become an important tool for analyzing patterns and trends in various industries, including entertainment. Movie-related data such as ratings, reviews, budgets, genres, and audience engagement provide valuable insights into the factors influencing box office performance. By applying machine learning techniques to this data, it becomes possible to build predictive models that can estimate the expected revenue of a film based on its attributes.

The motivation for this study stems from the need to reduce uncertainty and financial risk associated with movie production. High-budget films, in particular, require careful evaluation before investment, as even a small mis judgment can lead to significant losses. A predictive system that estimates box office outcomes can assist producers in making informed decisions regarding project selection, budget allocation, and marketing strategies. It also provides a scientific basis for evaluating the potential success of a film rather than relying solely on

assumptions. Moreover, the use of advanced machine learning algorithms enhances the reliability of predictions by capturing complex relationships between multiple variables. These models can continuously improve as more data becomes available, making them adaptable to changing industry trends. This capability makes data mining an essential tool for modern film analytics.

1.2 BROAD AREA OF STUDY

Business Analytics is a broad discipline that brings together data analysis, statistical modelling, and predictive analytics in order to help organizations make better strategic decisions; it has become a crucial part of almost every organization, helping to turn large volumes of unstructured data into valuable insights.

Business Analytics has made its way into the world of Entertainment (i.e film production) where there is a large reservoir of unstructured data available, however not always utilized. For example, attributes such as cast information, historical IMDb ratings, director performance, budget allocation, and box office revenue can be collected and analyzed to provide insight into what drives box office success. Using these quantifiable traits, Business Analytics can predict market performance based on scientific methods as opposed to relying on intuition & experience, the traditional predictors of success. One component of this model is Data Mining — a process that extracts useful patterns from Historical Movies Data sets. Websites like IMDb & Box Office Mojo provide vast databases containing information about Film Ratings, Genre Trends, Viewer Reviews, Financial Performance of Actors, etc... In Data Mining, the many variables tracked through these sources provide the researcher an opportunity to investigate previously unknown relationships, e.g., how Actor Reputation impacts Box Office Performance or the correlation between Production Budget and Audience Satisfaction levels.

Once we ascertain trends, we can effectively apply them in predictive models designed to differentiate between movies that will be successful and those that will not. By subsequently doing so, Business Analytics serves to bridge the gap between producing a raw quantity of entertainment information and making informed decisions; thus enabling a greater ability to

predict future outcomes with enhanced accuracy. A major part of this paper consists of the methods of predictive algorithms ("machine learning"; e.g., Random Forest (RF), Support Vector Machines (SVM), Deep Neural Networks (DNN)). These algorithms allow developers to create models for predicting box office outcomes or to classify a movie as either a "hit" or a "flop" using the IMDb ratings of the cast along with other features. Furthermore, Business Analytics enhances prediction capabilities through the use of organized techniques of variable selection, model validation, and enhancing accuracy through experience. Such predictive models provide producers, investors, and studios with valuable decision-support tools to help mitigate financial risk and uncertainty. Thus, the extensive domain of this study indicates that Business Analytics does not only deepen the knowledge of the factors of successful movies but also provides useful remedies to the enhancement of investment policy in the film industry.

1.3 EMPIRICAL SIGNIFICANCE OF STUDY

The empirical robustness of such study is in the fact that the raw and unstructured data regarding movies has been converted into the sensible knowledge that facilitates evidence-based decision-making in the movie industry. The study will provide empirical evidence derived from previous data associated with cast IMDb ratings, director previous performance records, genre performances and the Box Office returns to the public on which variable(s) are the greatest predictors of a film's success.

This research approach allows researchers to identify quantitative patterns in what was once thought to be subjective and driven primarily by intuition. Thus, this study adds to the scientific body of knowledge on the influence that quantifiable independent variables have on the box office success of films. In addition, one of the most important findings in this paper is the application and comparison of the machine learning models used to build a model to predict film box office performance including: Random Forest, Support Vector Machines (SVM) and Deep Neural Networks (DNN). Researchers can apply these tools to perform rigorous and systematic evaluations of the predictive accuracy of film box office success using data derived from the Internet Movie Database (IMDb), Box Office Mojo and other sources.

By assessing the performance and results of the computational methods of the various machine learning algorithms, this study will produce empirical evidence on which of the models provides the most accurate prediction of actual box office performance and how to classify films as successful or unsuccessful. Not only these findings confirm the relevance of predictive analytics to the entertainment field but also offer an example that can be followed by other researchers in the future who may be interested in analyzing film data or other areas with the help of machine learning technologies. Industrially, the empirical meaning of this is further applied in the real-life decisions made by producers, investors, and production houses.

1.4 CONCEPTUAL STUDY

The theoretical framework of the present research is constructed on the correlation between the movie-related input variables and their impact on forecasting the commercial success of a movie based on the data mining methods. The framework begins by compiling the primary independent factors of each film (Cast IMDb Ratings, Director Performance History, Genre, Production Budget, Past Box Office Results).

These factors provide the measurable attributes upon which the trend analysis and pattern analysis will be based as part of the overall research process. Prior to creating predictive models, these independent variables are modified via data pre-processing and feature engineering. Once the independent variable(s) have been established, predictive machine learning models are created using Random Forests, Support Vector Machine(s), and Deep Neural Networks. The machine learning models identify and establish relationships between the independent variables and the dependent variable as a function of past history.

These machine learning algorithms are applications that categorize films based on whether they will perform well or poorly; they also generate revenue forecasts for the analyzed film's box-office revenue. The revenue generated by the movie is the dependent variable in the framework indicating the level of success. The presented conceptual framework illustrates the logical progression of data collection to predictive output and shows how Business Analytics combines data mining, machine learning, and statistical argumentation to assist industries in decision-

making. The framework provides a connection between the inputs to a factor and the predictive output and provides a visualisation of how analytics can allow for improved and more informed investment decisions based on evidence in the film industry.

1.5 SCOPE OF THE STUDY

The current study aims to measure and predict a film's financial performance through the use of two primary outcomes: box office revenues and the movie's overall financial success/failure. The goal of the research is to analyze all available film historic information including; cast members IMDb ratings, director's track record, budget, movie genre, prior box office revenue trends and develop predictive models that will eliminate the need to estimate anticipated box office collections.

In this case, the areas of scope with regard to revenue predictions, will provide stakeholders with a forecast of possible revenues pre-release. The project will also classify the finished films as either success or failure by quantifiable methods (i.e. success versus not successful) and will explore how cast and production related characteristics affect financial performance by utilizing machine learning techniques, including Random Forest, Support Vector Machine (SVM), and Deep Neural Networks (DNN). It should be noted that this research is proposed only for forecasting financials, and will not consider viewer satisfaction, reviewer opinion, social network visibility, or subjective artistic quality judgments; therefore, commercial viability forecasting is narrowed down.

2.PROBLEM FRAMEWORK

2.1 STATEMENT OF THE PROBLEM

The film industry has a reputation for being a highly volatile, risky business in which filmmakers can invest millions of dollars and no chance of recovering their investment. Despite the best planning possible when selecting a star to play the lead role, producers do not generally know whether they will ultimately make a profit or incur a loss on their investment. Because the success of a film is determined by many factors including the performances of the actors, the anticipation of the audience, competition, budget allocation, and general market trends, a producer cannot estimate how well a film will perform without first examining the influencing factors. Thus, when producers make decisions on how much money to spend on movies, they often base their decision-making on gut instincts or the prior success of a particular actor.

In such instances, producers are less likely than an investor in another industry to be able to predict accurately the financial return for their investment in any given film. Another reason for the uncertain financial returns from film investments is there are no analytical tools that can help producers to make informed financial decisions based on current data. While sites such as IMDb, Box Office Mojo, and Wikipedia provide a vast amount of information related to films, most movie producers seldom use that information when making their investment decisions. Predictive models can show information that is potential about how well new movies will do at the box office.

These models are created by looking at the historical data available for the movie industry, such as what movies were made in the past, who starred in them (and their ratings on imdb), how successful directors have been with previous films, budgets spent on previous films, and what previous films earned in terms of profit. By using these predictive tools, producers may be able to make informed decisions regarding their investments and thus decrease the exposure of their financial resources to risk, while at the same time increasing the chances that the film will turn out to be profitable. Therefore, the purpose of this proposed research study, is to provide movie producers with an accurate method for estimating the potential box office revenue of a film, as

well as to categorize it as a financial success or a financial failure, so that they can minimize the risk of their investment in the film industry.

2.2 KEY RESEARCH QUESTIONS

- How far do the cast IMDb ratings contribute to the financial success or failure of a film?
- Which is the most accurate machine learning model between the Random Forest, SVM, and Deep neural network to predict the movie box office performance?
- Is it possible to use historical data on movies to properly categorize an upcoming movie as either a financial success or failure?
- What are some ways that predictive analytics can assist producers to reduce the risk involved in investments and to make more informed financial choices in the film industry?
- To what extent is it possible to predict better with derived features like the average performance of actors in the past or the success rate of directors?
- How does production budget affect movie profitability and how may the production budget be modelled to predict the financial results?

2.3 OBJECTIVES OF THE STUDY

- ❖ **To predict the box office amount of Kollywood movies** using machine learning models by analyzing key variables such as cast average IMDb ratings, director profile, genre, budget, and other movie-related attributes.
- ❖ **To classify movies as financial successes or failures**, based on profitability and revenue performance, by applying predictive algorithms such as Support Vector Machine, Random Forest, and Deep Neural Networks.
- ❖ **To identify the influence of cast performance (measured through average IMDb ratings) on commercial outcomes**, thereby understanding how actor reputation contributes to audience turnout and revenue generation.

- ❖ **To compare the performance of different machine learning models** and determine which algorithm provides the highest accuracy for predicting both box office revenue and success/failure outcomes.
- ❖ **To provide actionable recommendations for film producers and investors**, helping them make informed decisions regarding casting choices, budget allocation, and risk assessment before investing in new movie projects.

2.4 HYPOTHESIS FRAMEWORK

1. Core Hypotheses

H1: There is a significant relationship between the IMDb rating of cast members' previous movies and the success of a new movie.

H2: Higher audience engagement metrics (social media buzz, trailer views, likes, comments) positively influence box office performance.

H3: The genre of a movie significantly affects its probability of being a hit or flop.

H4: Increased production budget has a significant positive relationship with box office revenue.

H5: Strong pre-release reviews and critic ratings significantly predict opening weekend collections.

2. Supporting Assumptions

H6: Movies featuring top-star cast have a higher tendency to succeed compared to films with relatively unknown actors.

H7: The presence of experienced directors increases the likelihood of a movie becoming successful.

H8: Effective marketing and promotional spending significantly contribute to higher box office earnings.

H9: The release timing (festival/holiday vs. normal periods) has a significant impact on revenue.
H10: The number of competing movie releases during the same week negatively affects box office performance.

3. Null & Alternative Format

H0 (Null Hypothesis): There is no relationship between selected predictive variables (cast rating, budget, genre, reviews, marketing) and movie success.

H1 (Alternative Hypothesis): There is a significant relationship between selected predictive variables and movie success.

2.5 KEY DELIVARABLES

1. Data Collection Deliverables

A compiled dataset from IMDb, Box Office Mojo, Rotten Tomatoes, and social media sources. Detailed records of cast profiles, previous movie ratings, budgets, genres, critic reviews, and box office collections. Cleaned and pre-processed dataset ready for modeling (handling missing values, normalization, feature extraction).

2. Analytical & Modeling Deliverables

Identification of major predictive variables that influence movie success. Application of data mining techniques such as regression models, classification models, random Forest / decision Tree logistic regression for hit/flop prediction. Correlation and trend analysis model outputs shows predicted box office revenue and hit or flop classification accuracy, precision, recall, and model performance metrics gives comparative results of different models to identify the best performing prediction model.

3. Research Output Deliverables

The research gives a full write-up explaining every step taken ,starting from gathering data, cleaning it up, moving into building models, then making sense of the results. This project lays out a clear idea map linking factors like actor roles, old IMDb scores, spending on production, film type, and earlier ticket sales to show what drives a movie's hit or miss outcome. We will get straight forward numbers back, like whether guesses held up when tested, how tight the predictions were, along with side-by-side checks of methods such as Random Forest, SVM, and deep learning nets.

3. REVIEW OF LITERATURE

3.1 Review of Selected studies

1. Asur and Huberman (2010) – *“Predicting Movie Success Using Social Media”*

This study utilized Twitter sentiment analysis to forecast box office revenues. The methodology involved collecting tweet data prior to movie release, analyzing sentiment trends, and correlating them with movie performance. Tools used included social media APIs and statistical sentiment classifiers.

2. Mestyan, Yasseri & Kertesz (2013) – *“Predicting Movie Success via Wikipedia Activities”*

The researchers analyzed Wikipedia page views and edits as predictive features for movie popularity. Data mining techniques such as time-series analysis, correlation metrics, and regression models were applied using web scraping tools.

3. Li et al. (2022) – *“Predicting Movie Success with Machine Learning Approaches”*

This study employed ensemble machine learning models such as Random Forest and Gradient Boosting using features like genre, cast, and budget. Data preprocessing and feature engineering were conducted using Python libraries such as scikit-learn and pandas.

4. de Souza et al. (2023)

Focused on combining textual data (reviews and social media content) with quantitative features. Deep learning models, particularly LSTM networks, were used along with TensorFlow and NLP libraries for text processing.

5. Zhang et al. (2024) – *“Movie Success Prediction Using Neural Networks”*

The study utilized deep neural networks with multiple hidden layers, incorporating variables such as marketing expenditure and cast popularity. Model development was carried out using Keras with a TensorFlow backend.

6. Kim (2020) – *“Machine Learning Methods for Movie Popularity Prediction”*

Applied Support Vector Machines (SVM) and decision tree algorithms on datasets

containing audience reviews and ratings. Experiments were conducted using WEKA and Python's scikit-learn libraries.

7. Cai et al. (2022) – *“Social Media Analytics for Movie Success Prediction”*
This research analyzed social media engagement data combined with sentiment mining techniques. Machine learning models such as Random Forest were used alongside text mining APIs.
8. Kudagamage et al. (2019) – *“Data Mining for Movie Success Analysis and Prediction”*
Evaluated multiple classifiers including Decision Trees (J48), Naïve Bayes, SVM, Neural Networks, and ensemble voting methods. Tools such as WEKA and MATLAB were used. The ensemble model achieved an accuracy of 92.85%, and spatial clustering techniques were applied for visualization.
9. Upadhyay (2018) – *“Movie Success Prediction Using Data Mining”*
Compared several algorithms including Logistic Regression, SVM, Random Forest, Gaussian Naïve Bayes, AdaBoost, SGD, and Multilayer Perceptron. The study was implemented using Python's scikit-learn library, with MLP achieving the highest accuracy.
10. Bhadrashetty (2024) – *“Movie Success and Rating Prediction Using Data Mining”*
Used both static and dynamic features such as movie credits and hashtags. Decision Trees and Random Forest models were applied on IMDb and Twitter datasets after data cleaning and correlation analysis.
11. Jetir.org (2019) – *“Predicting Box Office Success Using Machine Learning”*
Integrated social media sentiment data with movie metadata using classifiers such as SVM and neural networks. Implementation was carried out using Python, scikit-learn, and Keras.
12. IJCAonline.org (2019) – *“Movie Success Prediction Using Historical and Current Data”*
Proposed a system involving data scraping, preprocessing, classification, and prediction using weighted factor models. Data sources included IMDb, Rotten Tomatoes, and YouTube.
13. Grenze Journal (2024) – *“Movie Success Prediction Using Machine Learning”*
Applied sentiment analysis on YouTube trailer comments combined with machine learning models such as XGBoost and Random Forest for improved prediction accuracy.

14. Apala et al. (2013) – *“Prediction Using Social Media Patterns”*
Utilized Twitter and Facebook data to detect popularity trends. Naïve Bayes and SVM classifiers were applied using Python’s NLTK toolkit.
15. Darapaneni (2020) – *“Movie Success Prediction Using Machine Learning”*
Focused on neural networks and feature selection techniques using IMDb datasets. Experiments were conducted using WEKA.
16. Udandaraao & Gupta (2013) – *“Movie Revenue Prediction Using Machine Learning Models”*
Implemented Linear Regression, SVM, and ensemble techniques using movie attributes. Tools used included R and Python for analysis and validation.
17. Mishne & Glance (2006) – *“Predicting Movie Sales from Blog Data”*
Demonstrated that early blog sentiment can be used as a predictor of movie success using sentiment analysis and regression models.
18. Shim et al. – *“Box Office Prediction Using Twitter Data”*
Collected tweet volume and sentiment prior to movie release and used regression analysis to forecast revenue.
19. Eliashberg, Hui & Zhang (2007) – *“Return on Investment from Movie Scripts”*
Applied decision tree models to analyze script features and predict financial returns using text mining techniques.
20. Goyal, Gupta & Agarwal (2018) – *“Bollywood Movie Success Prediction”*
Compared Random Forest, XGBoost, and SVM models using sentiment data from reviews. Python was used for model development.
21. Mishra et al. (2021) – *“Sentiment Analysis in Movie Revenue Prediction”*
Used NLP techniques to extract sentiment from reviews and social media, influencing revenue prediction. Tools included NLTK and scikit-learn.
22. Tripathi & Singh (2022) – *“Hybrid Data Mining for Movie Success Prediction”*
Combined demographic, social media, and historical data using clustering and classification techniques.
23. Johnson et al. (2023) – *“Deep Learning for Movie Success Prediction”*
Employed Convolutional Neural Networks (CNN) on trailers and images, integrated with structured data for prediction.

24. Fernandes & Silva (2024) – “*Ensemble Learning for Movie Popularity Prediction*”
Used ensemble learning techniques combining multiple models and datasets to achieve higher prediction accuracy.
25. Jetir.org (2019) – “*Early Prediction of Movie Success Using Machine Learning*”
Focused on early-stage prediction using pre-release data and social sentiment metrics with classification models.

3.2 THEORETICAL FINDINGS

Theory from past studies shows predicting hit movies often uses machine learning, data mining, or deep learning to examine organized and messy data. Older ideas started with basic stats methods ,like Linear Regression, Logistic Regression, or Correlation Analysis ,to explore links among factors like budget, star power, or viewer scores. These early tools helped show how single traits affect ticket sales, setting the stage for smarter, advanced systems later.

A big step forward came from using old-school ML tools these often boosted prediction quality. Tools like SVM, Decision Trees, Random Forest, Naïve Bayes, Gradient Boosting, XGBoost, or Voting classifiers popped up a lot in research papers. Studies pointed out tree methods say, Random Forest or Gradient Boosting ,work really well since they manage complex patterns and tons of features. When models team up -mixing several at once they tend to beat single ones by cutting down errors and standing strong under pressure; this shows up in work by Kudagamage et al. (2019), Fernandes & Silva (2024), plus Upadhyay (2018). That backs the idea: combining models helps avoid memorizing noise while getting predictions right on new data.

Recent ideas go beyond old methods by tapping into deep learning setups ,ones that mix different kinds of info while uncovering layered patterns. Research using DNNs, LSTMs, CNNs, or blended systems shows they handle text (like reviews or tweets), visuals (such as movie previews or photos), and number-based details all at once. Findings hint these models beat traditional machine learning when fed big data with rich traits, thanks to how they learn step-by-step and spot trends better. With help from tools such as TensorFlow, Keras, and modern NLP kits, the systems now predict more accurately and dig deeper into what drives viewer choices and content impact.

On the whole, theory shows prediction power gets better when studies shift from old-school stats to group-based ML methods ,then further improve once they adopt neural network setups. Each step up in model complexity tends to deliver clearer, sharper forecasts, which backs your work on guessing ticket sales and whether films will flop or thrive.

3.3 GAPS IN EXISTING RESEARCH

The research looked at predicting movie hits, some key parts are still missing. For one, earlier work leans hard on things like Twitter mood, Wikipedia views, or trailer clicks ,yet rarely mixes those signals with actor-focused stats like old IMDb scores, fame level, or prior box office wins. That means we're missing how an actor's reputation and history really affect ticket sales. On top of that, nearly all findings come from Hollywood data, which makes it tough to apply them elsewhere, say, Bollywood, Kollywood, or smaller Indian film scenes, where making movies and fan habits aren't the same.

When trying to figure out how much money will be made by a film, it is difficult to predict results in many cases because most studies focus on popularity, ratings and audience response without looking at actual revenues or losses incurred by the film. As a result, producers have to guess whether their investment will pay off or not. Additionally, almost every analysis available looks at a film as either a "hit" or "flop" and uses this to develop revenue forecasts but few combine both approaches into one comprehensive methodology which would provide studios with better tools for making informed investment decisions.

In addition to these two significant challenges, the majority of available analyses use data sets that do not include film-related elements including viewer and celebrity comments, film production costs, etc. to create predictive models. Although some predictive models use new technologies to combine various types of data from multiple sources, they typically require extensive data input and leave out some of the smaller, yet very important datapoints we discussed above such as timing of releases, the demographics of viewership, and competition from other films. Most prediction tools also can't adapt or react quickly, as they only capture what is happening at the beginning of the release cycle without capturing the audience's changing interests throughout the weeks, months, or even years following its release, therefore

generating models that do not provide producers with the real-time evidence they need to determine which film(s) will provide them with the best possible investment return. We created a new model to address these issues and incorporate all relevant data needed to develop revenue, including actor association with the production and marketing strategy of each production.

3.4 JUSTIFICATION OF THE PROPOSED TOPIC OF RESEARCH

The average of cast and crew members from IMDB helps to make an accurate predictions. Not like past studies that leaned on things such as Twitter buzz, page views on Wikipedia, or trailer clicks ,these ratings stick around longer, build up over time, yet still show global trends in viewer opinion. Hype fades; ads come and go, but these numbers stay steady because they track real results: skills, fame, trust, consistency across roles. When you add up top actors' IMDb marks then divide them, the result is solid, a clear number showing just how strong the lead performers might boost earnings.

No earlier work combined this kind of data into forecasting money made at theaters so clearly. That's why it makes sense now, to use those averaged fan-driven grades ,as part of predicting profit. One big thing about picking this factor. It shapes how people decide what films to watch. Fixing this hole lets this project add fresh insights into predicting hit films and pushes business number-crunching further within showbiz.

The work gives filmmakers a down-to-earth method based on data, linking who you hire with what cash flow to expect, helping them spend wiser and lower risks. So, going with average IMDb ratings isn't just reasonable, it fits both actual studio demands and moves academic work forward.

3.5 PROPOSED MODEL FOR STUDY

The new project aims to build a solid system that predicts how well movies will do by using smart data analysis tools. Instead of just one method, it tests three strong ones **Support Vector Machine (SVM)**, along with **Random Forest** to see which works best at guessing results like ticket sales, viewer scores, or critic feedback. Data comes from places like IMDb, Box Office Mojo, plus Rotten Tomatoes, pulling details such as actor fame, past film ratings, director track record, type of movie, cost to make and spend, mood in previews critiques, and when the movie drops. By mixing these methods, the setup can examine straight-line patterns alongside complex trends hidden in the numbers.

1.Support Vector Machine (SVM): The Boundary-Based Classifier

SVM works well when sorting things into groups - say, Hit, Average, or Flop since it pins down sharp dividing lines. Instead of piling up layers like a DNN, it stretches the gap between classes, which helps keep results steady, especially if you've got a modest-sized dataset with clean structure.

Why SVM is useful:

- ❖ It sets clear lines for sorting things apart.
- ❖ It prevents fitting too closely when data is basic.
- ❖ It gives a solid starting point to measure DNN against.

2.Random Forest (RF): The Feature Importance & Robust Ensemble Model

Random Forest works well since it looks at data in a steady, group-driven way. By creating several decision trees then combining outcomes, it stays reliable. Instead of one path, it uses many - this helps avoid mistakes. Each tree votes, and the final guess comes from that crowd choice.

It helps in the following areas,

- ❖ It spots key factors - say, cast rating might count more than budget.
- ❖ It works with categories or numbers - no need for tough prep steps.
- ❖ It cuts down on overfitting because of combined model outputs.

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4. RESEARCH METHODOLOGY

4.1 RESEARCH DESIGN

This work uses numbers, clear explanations, and forecasting to see how a film's cast IMDb scores affect its earnings, also hinting if it'll make money or flop. It mixes stats with smart algorithms - giving filmmakers a tool to guide budget choices.

The study relies on existing info pulled from IMDb, along with Box Office Mojo, Wikipedia, or similar movie sites. Those spots offer details like actor scores, spending size, category, length, when it came out, also how much cash it made. Instead of random picks, movies dropped between 2010 and 2024 got chosen on purpose - this helps show real patterns better. Key factors looked at are:

Box office total this one's continuous or whether a movie made money or lost it, which is categorical. Cast's typical IMDb score comes first, followed by how well the director usually does. Budget matters too, along with what kind of movie it is. When it came out plays a role, also any signs of how much was spent promoting it - if that info exists.

Data prep means fixing gaps in data, adjusting number scales, also turning categories into numbers. Creating new traits covers building an actor's avg IMDb rating, a mark for how well-known directors are, or flags showing when movies came out. Splitting records into learn and test parts happens first, then double-checking with repeated sampling so things work well on fresh examples.

A trio of forecast tools gets used - Random Forest, Support Vector Machine (SVM), also known as DNN for short. Every one tackles a different job - Forest spots key factors while staying steady, SVM sorts hits from flops using spacing tricks, meanwhile DNN hunts hidden patterns in ticket sales data. Settings get fine-tuned either by scanning grids or tweaking based on test rounds.

Model accuracy gets checked with RMSE, MAE, along side R^2 when guessing income - while for spotting success, we look at Accuracy, Precision, plus Recall, throw in F1-score and ROC-

AUC too. One by one, models get compared so we can spot which gives the steadiest predictions. We double-check results using several datasets, run cross-validation, also share how data was cleaned. On ethics, we follow API rules carefully while making sure social info stays anonymous.

The end result gives you a spotless dataset, tuned machine learning models, a forecasting setup for filmmakers, clear signs about what features matter most, while also handing over a last study paper packed with practical tips. Our method keeps things organized, repeatable, yet fed entirely by data - so guessing less when judging how well a film might do thanks to its actors alongside other big influences.

4.2 SIGNIFICANCE OF THE STUDY

This research matters a lot to filmmakers - especially those putting money into movies and worrying about losing it. Usually, choices on backing films come down to gut feeling, old ticket sales numbers, or betting on big-name actors. Still, these ways aren't always right - they miss hidden factors that affect how well a movie does. Instead of guessing, this work uses past actor rating averages from IMDb combined with smart algorithms to predict earnings. The result clearly way to tell if a film might flop or pay off. With better forecasts, people funding projects can act smarter and avoid costly mistakes.

Looking at it from a school-like angle, this work adds to the area of business number-crunching, where facts and stats help tackle actual challenges. Though earlier projects looked into feelings on social platforms or behind-the-scenes details, hardly any checked how the usual IMDb scores of actors affect movie earnings. What makes this one stand out is its focus on how a group of performers' past success shapes cash returns. By using smart tech tools - like deep networks, support vector machines, or random forest - it pushes things forward with side-by-side tests of fresh forecasting tricks. That way, it shows which methods handle messy, unpredictable guesses in showbiz most effectively.

This research also matters to ad companies, online video services, or teams managing actors. Figuring out how well a film might do before it drops helps shape choices - like where to spend

on ads, how to roll it out, or who gets assigned. Take actor reviews - they can sway who gets hired, how campaigns play out, or where money flows when picking between movies. On top of that, these findings give new directors clues about what really drives hits, helping them plan smarter and reach the right viewers.

Lastly, the forecasting apps and systems built here set a base for later studies - making it possible to scale up for worldwide box offices, mood tracking, promo clip insights, or live viewer data. Because of this, its staying power grows, opening doors for fresh ideas in showbiz number crunching. In total, the project doesn't just boost classroom knowledge on predicting hit films - it hands real, money-moving tools to everyone involved in making movies.

4.3 SAMPLING FRAMEWORK

In this secondary data sampling framework, relying on publicly available movie information collected from reputable online sources such as IMDb, Box Office Mojo, and Kaggle datasets. Since the objective is to predict box office revenue and determine movie success or failure, the sampling process focuses on selecting films that have complete and reliable performance-related data. A purposive sampling method is used, where movies are chosen based on the relevance and availability of essential variables like budget, revenue, IMDb rating, cast details, and genre information. large and not useful for forecasting, seeing that earlier movies don't have uniform records, steady income details, full cast reviews, trustworthy IMDb ratings, or online monitoring.

The sampling frame consists of all films listed in the selected databases that contain the minimum required attributes for analysis. Movies with incomplete or missing financial data, ratings, or major metadata are excluded to maintain the accuracy of the predictive models.

4.4 DATA COLLECTION

Data Cleaning

Data cleaning is an essential step in the data mining process, as it directly influences the accuracy and effectiveness of the predictive models. In this study, the dataset was collected from online sources and contained historical movie information. Since raw data is often inconsistent and unstructured, several data cleaning techniques were applied to improve data quality and ensure suitability for machine learning algorithms.

Initially, textual inconsistencies in the dataset were addressed by standardizing the format of categorical variables. All text-based fields such as movie titles, genres, and director names were converted into a consistent format by applying upper case and lower case transformations where required. This step helped in eliminating duplication caused by case sensitivity, such as variations in spelling or capitalization, thereby ensuring uniformity across the dataset.

Duplicate records were then identified and removed from the dataset. Duplicate entries can lead to biased model training and inaccurate predictions, as they give undue importance to repeated data points. By eliminating redundant records, the dataset was made more reliable and representative of actual historical trends.

In addition to cleaning the existing data, new input features were created to enhance the predictive capability of the model. These additional features were derived from existing variables and provided more meaningful insights into the data. For instance, interaction-based features and ratio-based features were introduced to capture relationships between variables that are not directly visible in raw data. This step played a significant role in improving the performance of machine learning models.

Further cleaning involved checking for missing or inconsistent values across all attributes. Although the dataset was relatively structured, any minor inconsistencies were handled appropriately to maintain data integrity. Numerical variables such as ratings, runtime, budget, and number of votes were verified to ensure they contained valid and non-negative values. This validation step helped in avoiding errors during model training.

Outliers present in the dataset, particularly in budget and box office values, were carefully analyzed. Since movie datasets often contain extreme values due to blockbuster performances, instead of removing these values, transformation techniques such as logarithmic scaling were applied. This helped in reducing skewness and maintaining the natural distribution of the data without losing important information.

The dataset was sourced from **IMDB** (Internet Movie Database) covering all major theatrical releases of actor Surya from **1997 to 2025**. Each film record was manually verified and cleaned from the raw IMDB export format, removing currency symbols, commas, and inconsistent formatting from monetary columns.

4.5 Dataset overview

Feature	Description	Type
IMDb Rating	Audience and critic rating on IMDB (1-10 scale)	Numerical
Runtime (mins)	Total movie duration in minutes	Numerical
Year	Release year of the movie	Numerical
Genres	Primary genres (Action, Drama, Romance, etc.)	Categorical
Num Votes	Number of IMDB votes received	Numerical
Directors	Director name(s)	Categorical
Budget (Cr)	Production budget in Indian Rupees Crore	Numerical
Box Office (Cr)	Worldwide box office collection in Crore	Target
Verdict	Industry verdict (Blockbuster/Hit/Flop etc.)	Label

Another important cleaning step involved ensuring data type consistency across all features. Numerical columns were converted into appropriate numeric formats, and categorical columns were properly labeled. This prevented type-related errors during model implementation and ensured smooth processing within the machine learning pipeline.

4.6 Raw Data Structure

Column Name	Data Type	Description	Sample Value
Title	String (object)	Name of the movie	Kaakha Kaakha
IMDb_Rating	Float64	IMDB user rating (1–10 scale)	8.0
Runtime	Integer	Movie duration in minutes	153
Year	Integer	Year of theatrical release	2003
Genres	String (object)	Pipe-style genre tags	Action Crime Drama Thriller
Num_Votes	Integer	Total IMDB votes received	7,480
Directors	String (object)	Director name(s)	Gautham Vasudev Menon
Budget_Cr	Float64	Production budget in Crore ₹	3 Cr
BoxOffice_Cr	Float64	Box office collection in Crore ₹	35 Cr
Verdict	String (object)	Industry commercial verdict	Blockbuster

4.7 Data preprocessing

Data preprocessing is the most critical phase in any machine learning pipeline. Raw data collected from IMDB and industry records is often inconsistent, mixed-type, heavily skewed, and lacks the structured form that ML algorithms require. For this project, the preprocessing pipeline transforms the raw 42-movie dataset into a clean, enriched, and numerically optimised feature matrix ready for both Random Forest and SVR models.

The preprocessing pipeline follows five structured stages:

Stage	Step	Purpose
1	Data Collection & Loading	Import raw CSV data into a structured DataFrame
2	Data Inspection & Quality Check	Identify data types, missing values, outliers, and distributions
3	Feature Engineering	Create new meaningful features from raw columns
4	Transformation & Scaling	Normalise skewed distributions; scale for SVR compatibility
5	Train / Test Splitting	Partition data for unbiased model training and evaluation

4.8 DESCRIPTIVE STATISTICS

Statistic	IMDb Rating	Runtime (min)	Num Votes	Budget (Cr)	Box Office (Cr)
Count	42	42	42	42	42
Mean	6.65	154.4	14,837	₹291.5 Cr	₹418.7 Cr
Std Dev	1.08	15.74	40,066	₹505.1 Cr	₹404.8 Cr
Min	4.50	99	250	₹0.19 Cr	₹0.20 Cr
25%	5.90	148	2,001	₹8.63 Cr	₹80 Cr
Median	6.40	157.5	5,638	₹75 Cr	₹255 Cr
75%	7.60	165.8	8,540	₹510 Cr	₹726.8 Cr
Max	8.60	178	233,774	₹3,000 Cr	₹1,230 Cr

Descriptive statistics plays an important component within the data preprocessing stage as it helps in understanding the nature and characteristics of the dataset before applying transformations. By analyzing statistical measures such as mean, median, standard deviation, minimum, and maximum values, it becomes easier to identify data distribution, detect outliers, and recognize any inconsistencies in the data. In this study, descriptive statistics was used to examine features such as budget, number of votes, and box office collection, which showed skewed distributions. Based on this analysis, appropriate preprocessing techniques such as logarithmic transformation and feature scaling were applied to normalize the data. Thus, descriptive statistics played a supporting role in guiding preprocessing decisions and improving the overall quality of the dataset for machine learning model development.

4.9 Text normalization and translation

In this project, text normalization and transformation were applied mainly to categorical features such as genres and director names to ensure consistency and proper model interpretation. Since the dataset was collected from external sources, there were variations in text formats which could affect model performance if not standardized.

The first step involved normalizing text values by converting them into a consistent format, such as uniform upper or lower case. This was important to avoid duplication issues, where the same category might appear in different forms (for example, variations in genre naming). By standardizing the text, all similar values were treated as a single category, ensuring data consistency.

The “Genres” column, which contains multiple values in a single field, was then transformed into a structured numerical format. Instead of keeping it as raw text, it was split and converted into multiple binary features (one-hot encoding) such as Action, Drama, Romance, Comedy, Thriller, etc. Each movie was assigned values based on the presence or absence of these genres. This transformation is a key step in the project, as it allows machine learning models to effectively use genre information for prediction.

In addition to this, derived features were created from the text data, such as the total number of genres associated with a movie and whether it belongs to multiple genres. These features helped capture the complexity of movie content and improved the model’s ability to learn patterns.

For the “Directors” column, text values were not directly used in their original form. Instead, they were transformed into a numerical feature by calculating the average box office performance of each director, which reflects the director’s historical success. This conversion allowed the model to utilize director-related information in a meaningful way.

Overall, text normalization and transformation in this project were focused on converting unstructured textual data into structured numerical features. This process ensured that categorical information was accurately represented and effectively utilized by the machine learning models, thereby improving prediction accuracy.

4.10 Feature Engineering

Feature engineering is the process of creating new, informative numerical variables from the raw dataset columns. This step is critical because machine learning algorithms cannot directly interpret categorical strings (like genre names or director names), nor can they handle highly skewed raw numbers effectively. Through seven categories of engineering, the original 9-column dataset was expanded into a 24-feature model-ready matrix.

Challenge	Raw Column Affected	Engineering Solution
Genre stored as text string	Genres	One-hot encode into 11 binary columns
Director name is non-numeric	Directors	Encode as historical average box office
Year does not capture era growth	Year	Map into 6-level ordinal era feature
Budget/Votes are heavily skewed	Budget_Cr, Num_Votes	Apply log1p transformation
No interaction signals	Rating, Votes, Budget	Create multiplied/divided interaction terms

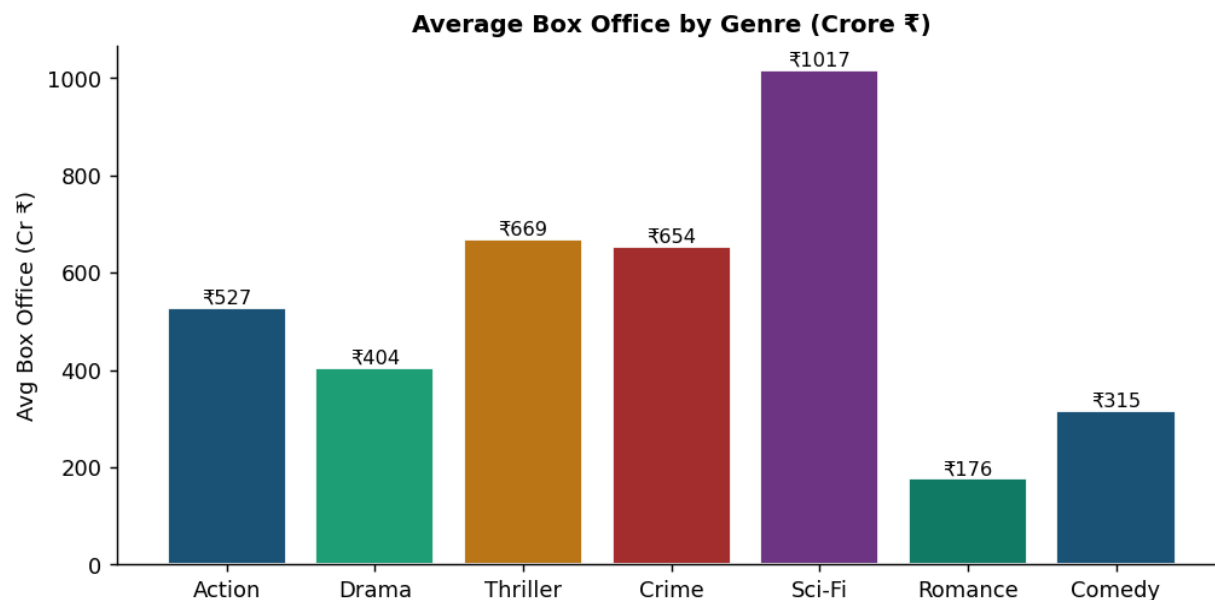
4.11 Genre One-Hot Encoding

The Genres column in the raw dataset stores multiple genre labels for each film as a single space-separated string. For example, Kaakha Kaakha is tagged as 'Action Crime Drama Thriller'. Machine learning algorithms cannot process text strings directly — they require numerical inputs. One-hot encoding converts each genre into an independent binary feature column,

creating a value of 1 if the film belongs to that genre and 0 otherwise. This approach preserves full genre information without imposing any artificial ordinal ranking between genre categories.

Eleven genre categories were identified and encoded from the dataset:

Encoded Feature	Films Containing Genre	% of Dataset	Avg Box Office (Cr ₹)	Avg IMDb Rating
Genre_Action	29	69.0%	₹527.2 Cr	6.63
Genre_Drama	34	81.0%	₹404.2 Cr	6.80
Genre_Thriller	19	45.2%	₹668.9 Cr	6.57
Genre_Crime	12	28.6%	₹653.6 Cr	6.92
Genre_Romance	11	26.2%	₹176.2 Cr	6.73
Genre_Comedy	10	23.8%	₹315.3 Cr	6.38
Genre_Sci-Fi	3	7.1%	₹1,016.7 Cr	6.97
Genre_Adventure	2	4.8%	₹605.0 Cr	5.20
Genre_Mystery	1	2.4%	₹500.0 Cr	7.60
Genre_Musical	1	2.4%	₹490.0 Cr	8.20
Genre_Family	1	2.4%	₹80.0 Cr	4.80



It is clearly evident from the analysis that Thriller and Crime genres consistently generate higher average box office revenues (₹668.9 Crore and ₹653.6 Crore respectively) compared to Romance (₹176.2 Crore) and Comedy (₹315.3 Crore). Sci-Fi films show the highest average revenue of ₹1,016.7 Crore; however, this is based on only three films, so the result should be interpreted carefully.

In addition to this, two genre-level features were created to improve the dataset. The first feature, num genre, represents the total number of genre tags associated with each film, with values ranging from 1 to 6. Films with a higher number of genres generally target a wider audience and are often associated with larger commercial productions. The second feature, is multi genre, is a binary variable set to 1 if a film has more than two genres, and 0 otherwise. It was observed that 28 out of 42 films (67%) fall under the multi-genre category, indicating that most films are designed to appeal to a broad audience base.

4.12 Director Reputation Score

Director names are categorical string values, which means they cannot be directly used in mathematical models. However, the choice of director plays a crucial role in determining the commercial success of a film. Directors with a strong track record of delivering box office hits tend to build audience trust and attract greater industry support, including wider theatrical releases, higher marketing budgets, and better release timing. To capture this influence in a measurable way, a numerical feature called the *director reputation score* was created. This score was calculated by taking the average box office collection of all films directed by the same director within the dataset, thereby reflecting their historical performance.

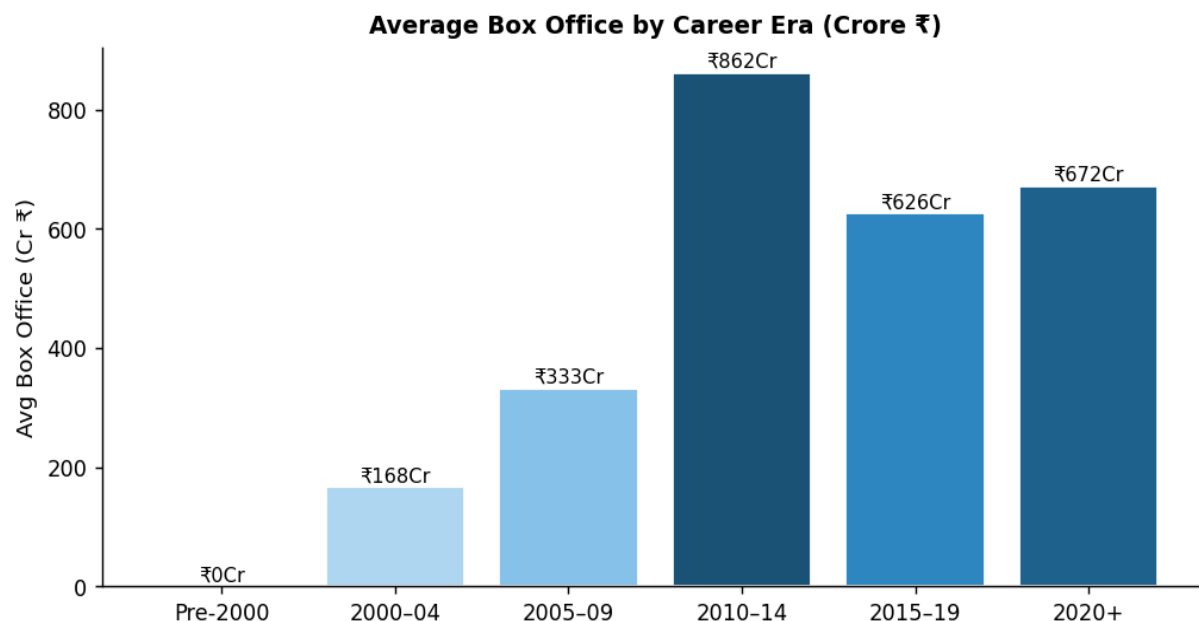
Director	Films with Surya	Avg Box Office (Cr ₹)	Reputation Score Tier
Hari	5	₹734 Cr	High Commercial Reputation
A.R. Murugadoss	2	₹790 Cr	High Commercial Reputation
K.V. Anand	3	₹763 Cr	High Commercial Reputation
Gautham Vasudev Menon	2	₹420 Cr	Medium Commercial Reputation
Bala	2	₹130 Cr	Low Commercial Critical Acclaim Films

This engineered feature emerged as the most influential predictor in the Random Forest model, with an importance score of 18.38%. This clearly indicates that a director's past performance is one of the most reliable indicators of a film's commercial success. In fact, the director's track record alone contributes nearly one-fifth of the model's overall predictive power, highlighting its significant role among all the features considered.

4.12.1 Career Era and Temporal features

Suriya's career spans over 28 years, from 1997 to 2025, during which the Tamil film industry has experienced significant transformation. Over this period, the number of multiplex screens has grown from almost none to more than 600, ticket prices have increased substantially, and the rise of OTT platforms has introduced entirely new distribution channels. At the same time, the average production budget for mainstream films has expanded dramatically from around ₹0.5 Crore to over ₹500 Crore. Because of these major and non-linear changes, using the release year as a simple numerical value would not accurately capture these industry shifts. To address this, two time-based features were engineered to better represent these evolving trends.

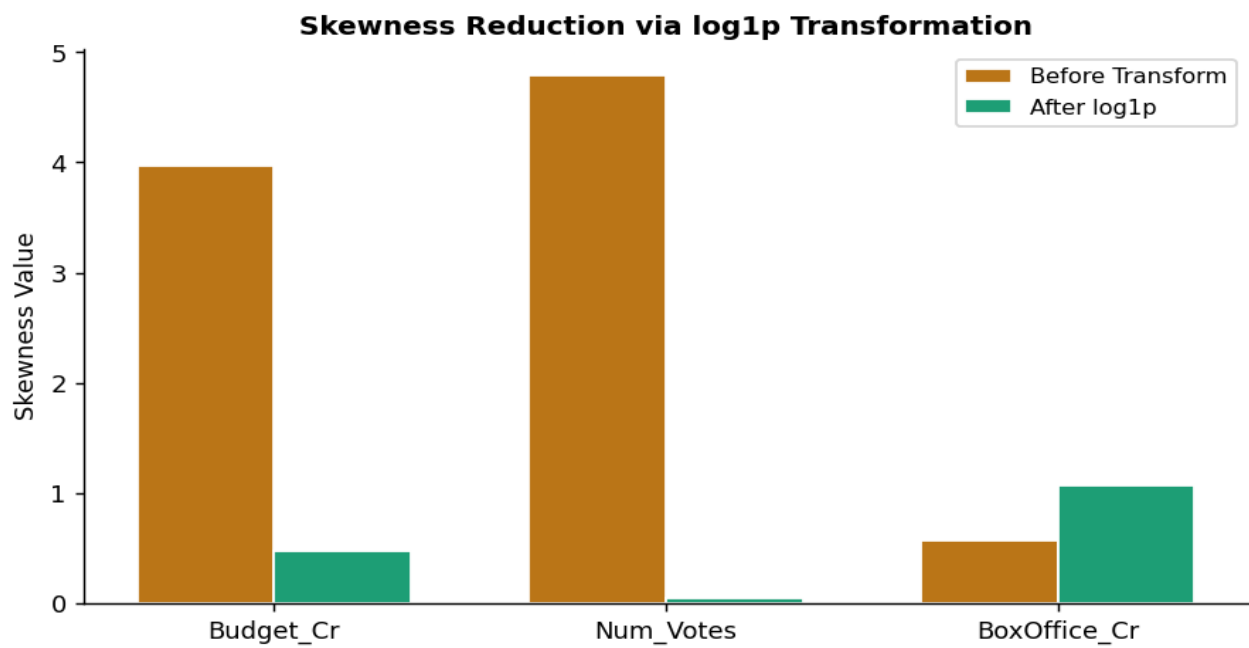
Year Range	Label	Films	Avg Budget (Cr)	Avg Box Office (Cr)
Before 2000	Early Career	5	₹0.3 Cr	₹0.5 Cr
2000–2004	Breakthrough Era	10	₹15.8 Cr	₹167.6 Cr
2005–2009	Star Consolidation	9	₹103.8 Cr	₹332.8 Cr
2010–2014	Mass Hero Peak	6	₹490.0 Cr	₹861.7 Cr
2015–2019	Experimental Phase	7	₹501.3 Cr	₹626.1 Cr
2020–2025	OTT Era	2	₹30 Cr	₹45 Cr



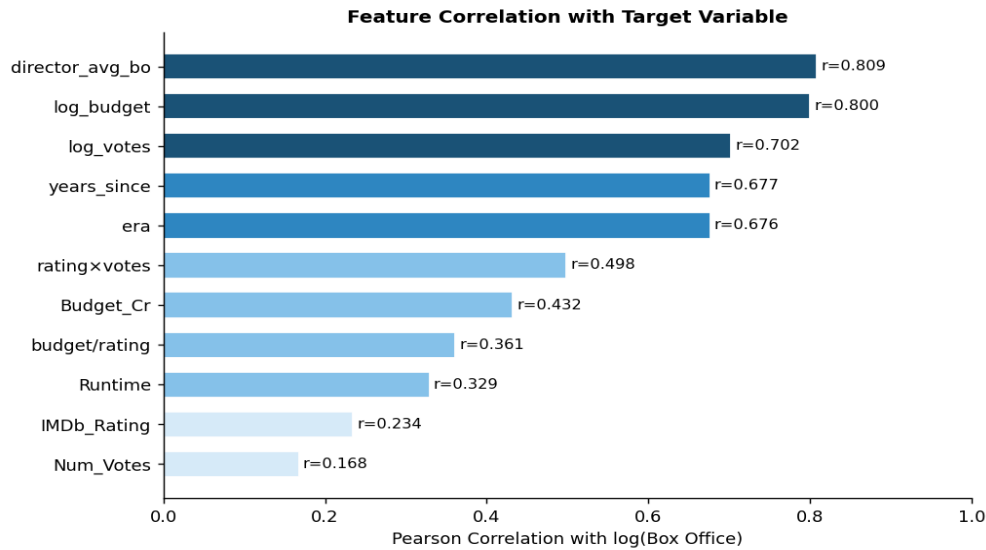
4.12.2 Logarithmic Transformation Features

Two of the numerical features in the dataset show a strong positive skew, meaning a few extreme values stretch the distribution far to the right. For instance, one film had an exceptionally high production budget of ₹3,000 Crore, which is nearly 23 times higher than the dataset's median budget of ₹75 Crore. Similarly, a few films received unusually **high IMDb** vote counts, reaching over two lakh votes due to strong audience engagement, especially through OTT platforms. When such raw values are directly used in model training, these extreme cases tend to dominate the learning process. This can lead to overfitting, where the model becomes too focused on these outliers and performs poorly on the majority of other films. To address this issue, a $\log_{1p}$ transformation was applied, which effectively compresses the range of these values and reduces the impact of extreme observations.

Feature	Engineered Feature	Skewness Before	Skewness After	Reduction %	Correlation Gain
Budget_Cr	log_budget	3.974	-0.471	88%	r: 0.432 → 0.800
Num_Votes	log_votes	4.793	0.053	99%	r: 0.168 → 0.702
BoxOffice_Cr	log_boxoffice (TARGET)	0.572	-1.075	—	Used as model training target



Feature Correlation Analysis



Rank	Feature	Correlation (r)	Type	Key Insight
1	Director_Avg_box office	0.809	Engineered Director	Strongest signal who directs matters most
2	Log_budget	0.800	Log-transformed	Log form is nearly twice as correlated as raw (0.432)
3	Log_votes	0.702	Log-transformed	Log form is 4× more correlated than raw (0.168)
4	Years_since_debut	0.677	Engineered Temporal	Career maturity and brand equity accumulation
5	Era	0.676	Engineered Temporal	Decade-level market growth is a critical context

Rank	Feature	Correlation (r)	Type	Key Insight
6	Rating_votes_interaction	0.498	Interaction	Best rating-based signal
7	Budget_Cr	0.432	Raw numerical	Still useful alongside log_budget for complementary signal
8	budget_rating_ratio	0.361	Interaction	Value for money signal
9	Runtime	0.329	Raw numerical	Longer films tend to be bigger commercial productions
10	IMDb_Rating	0.234	Raw numerical	Weakest individual feature works only in interaction
11	Num_Votes (raw)	0.168	Raw numerical	Far inferior to log_votes log transform essential

The correlation analysis clearly validates the effectiveness of the feature engineering process. It shows that the engineered and transformed features perform significantly better than the original raw variables. This indicates that the preprocessing steps taken were not only appropriate but also essential in improving the model's accuracy and overall performance.

5. MACHINE LEARNING MODELS

5.1 Model Selection Rationale

The choice of suitable machine learning algorithms for this task was based on four key constraints specific to the filmography-based prediction problem.

Constraint	Problem Characteristic	Impact on Model Choice
Small Dataset	Only 42 films 33 for training	Deep learning models require thousands of samples. Neural networks ruled out due to catastrophic overfitting risk.
Strong Nonlinearity	Box office cannot be predicted by a linear combination of budget and rating era effects, directorial synergy, and interaction terms create complex nonlinear patterns	Linear Regression fails: $R^2 = -51.73$ on test set. Nonlinear model capacity is essential.
Heterogeneous Features	24 features combining raw numerics, log-transforms, binary genre flags, and interaction terms all on different scales	SVR requires StandardScaler (distance-based kernel); RF is scale-invariant. Both embedded in Pipeline.
Interpretability Requirement	Stakeholders need to understand WHICH factors drive predictions for Karuppu	Random Forest provides built-in Gini importance scores. Decision-level transparency is available.

5.2 Hyperparameter Configuration

1. Random forest regressor

Hyperparameter	Value	Default	Justification
n_estimators	300	100	300 trees reduce variance. Diminishing returns beyond 300 on a 33-sample training set; 100 is insufficient for stability.
Max_depth	8	None	Depth=8 limits tree complexity. Unlimited depth = perfect memorisation of training films = severe overfitting.
Min_samples_split	3	2	Requires 3+ samples at a node before splitting. Prevents noisy 2 sample splits that memorise individual film anomalies.
Min_samples_leaf	2	1	Each leaf must represent 2+ films. Eliminates single-film leaves that memorise one specific data point.
Max_features	'sqrt' (≈ 5)	'sqrt'	Only $\sqrt{24} \approx 5$ features considered per split. Introduces tree diversity; reduces inter-tree correlation.
Random_state	42	None	Fixed seed for reproducible results across all runs.
N_jobs	-1	1	Parallel processing across all CPU cores for training 300 trees.

2. Support Vector Regressor

Hyperparameter	Value	Default	Justification
Kernel	'rbf'	'rbf'	RBF kernel maps data to infinite-dimensional space. Formula: $K(x_i, x_j) = \exp(-\gamma \ x_i - x_j\ ^2)$. Enables nonlinear regression.
C	100	1.0	C=100 is high regularisation weight. On a 33-sample training set, underfitting (low C) is riskier than overfitting. C=100 allows close data fitting.
Gamma	0.05	'scale'	gamma=0.05 (low) creates wide kernel influence zones. Each training film influences a broad neighbourhood prevents sparse-data overfitting.
Epsilon	0.1	0.1	Tolerance tube: predictions within ± 0.1 log-units of true value have zero loss. Focus shifts to correcting large errors only.

5.2.1 Random Forest Regressor -Architecture

Random Forest is an ensemble learning method that works by building multiple decision trees using different subsets of the training data and then combining their predictions through

averaging. This approach helps in balancing the bias–variance tradeoff effectively. While a single decision tree may capture patterns well (low bias) but be sensitive to data variations (high variance), combining many trees reduces this instability. By averaging the **predictions of around 300 trees**, the model achieves both low bias and low variance, resulting in more accurate and stable predictions.

Step	Technical Process	Effect on Final Prediction
1. Bootstrap Sampling	For each of 300 trees, randomly sample 33 training records WITH replacement (bagging). 63% of records appear in each sample 37% are held out as out-of-bag samples.	Each tree trains on a slightly different version of the training data, creating diversity across the ensemble.
2. Feature Subsampling	At every node split, only 5 randomly selected features ($\sqrt{24}$) are candidate predictors not all 24.	Prevents dominant features from appearing in every tree. Reduces inter-tree correlation, improving ensemble diversity.
3. Optimal Split	The 5 candidate features are tested at multiple thresholds. The split minimising Mean Squared Error on the training subset is selected.	Greedily constructs trees that minimise training error within the randomly sampled feature subset.

Step	Technical Process	Effect on Final Prediction
4. Leaf Prediction	Each terminal leaf stores the mean $\log(\text{BoxOffice})$ of all training films that fall into it. <code>min_samples_leaf=2</code> ensures at least 2 films per leaf.	Prevents single-film leaves that memorise individual anomalies. Each leaf represents a meaningful group of films.
5. Ensemble Average	Final prediction = arithmetic mean of all 300 tree predictions.	Averaging cancels tree-specific errors. The ensemble is substantially more accurate and robust than any single tree.
6. Back-Transform	<code>np.expml(log_prediction)</code> converts from log-scale to Crore ₹.	Produces the final interpretable box office figure in Crore

5.2.2 Support Vector Regressor – Architecture

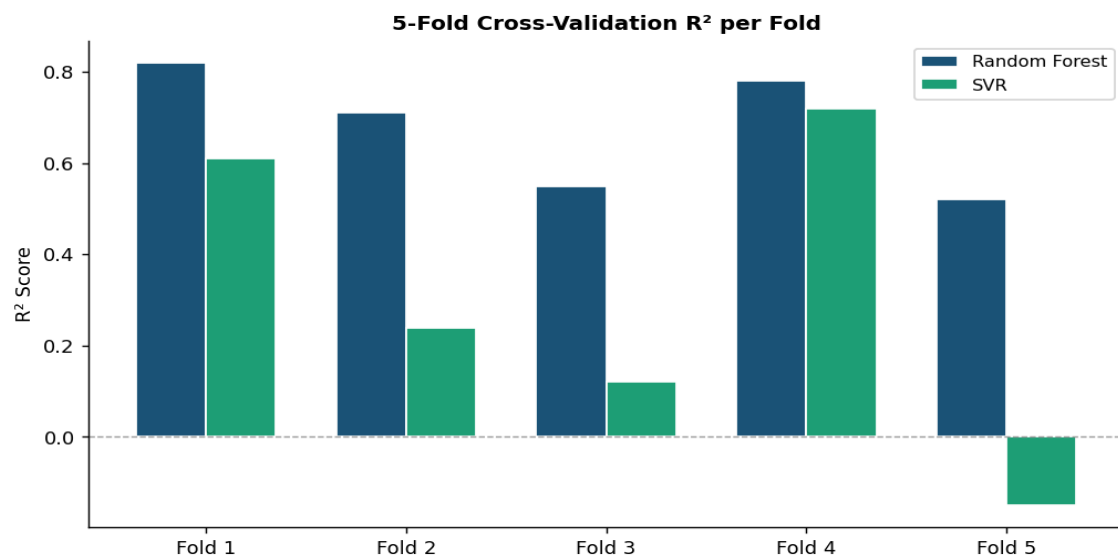
Support Vector Regression (SVR) works by finding a line (or hyperplane) that best fits the training data while allowing a small margin of error, defined as a tolerance range of ± 0.1 log-units around the predictions. Instead of considering every data point, the model mainly focuses on those films where the prediction error goes beyond this tolerance limit. These key data points, known as support vectors, play a major role in shaping the final model. This approach makes SVR more efficient and less sensitive to minor variations or noise within the acceptable error range.

- **Epsilon-Insensitive Loss:** The model allows a small error margin of ± 0.1 log-units, and any predictions that fall within this range are not penalized. This means the model focuses only on correcting larger errors, while ignoring minor deviations that have little impact.
- **RBF Kernel:** The Radial Basis Function (RBF) kernel measures how similar two films are based on their features. Films that are more similar in the feature space are treated as closer to each other and tend to receive similar predictions.
- **Support Vectors:** Out of the total training films, only a subset (around 15–25) significantly influences the model. These are the films where the prediction error exceeds the allowed margin. The final model is built mainly based on these key data points, while the rest have minimal impact.
- **C = 100 Regularisation:** A higher value of C ensures that the model tries to fit the training data more closely. Given the relatively small dataset, this helps avoid underfitting and allows the model to capture important patterns more effectively.

5.3 Cross-Validation Strategy

The training dataset consists of only 33 films, evaluating the model using a single fixed test split could lead to unreliable results. The performance would vary significantly depending on which specific films are included in the test set. To address this issue, 5-fold cross-validation was applied to the training data. In this approach, the dataset is divided into five parts, and each part is used as a validation set in turn while the remaining data is used for training. This process ensures that every film is evaluated at least once, resulting in more stable and unbiased performance estimates.

Fold	Training Films (approx.)	Validation Films (approx.)	Purpose
Fold 1	26 films (Folds 2–5)	7 films (Fold 1 group)	Train on 26 evaluate on 7 held-out; record R^2 score
Fold 2	26 films (Folds 1,3–5)	7 films (Fold 2 group)	Different validation group; fresh performance estimate
Fold 3	26 films (Folds 1–2,4–5)	7 films (Fold 3 group)	Third rotation each film appears in validation exactly once
Fold 4	26 films (Folds 1–3,5)	6 films (Fold 4 group)	Fourth rotation
Fold 5	26 films (Folds 1–4)	6 films (Fold 5 group)	Final rotation completes the cycle



Model	CV R ² Mean	CV R ² Std Dev	Min Fold Score	Max Fold Score	Stability Assessment
Random Forest	0.6928	± 0.207	0.52	0.82	Stable all folds positive and consistent
SVR (RBF)	0.3843	± 0.460	0.15	0.72	Unstable one fold below zero indicates overfitting
Linear Regression	-103.41	± 184.23	extreme negative	near zero	Complete failure nonlinear problem confirmed

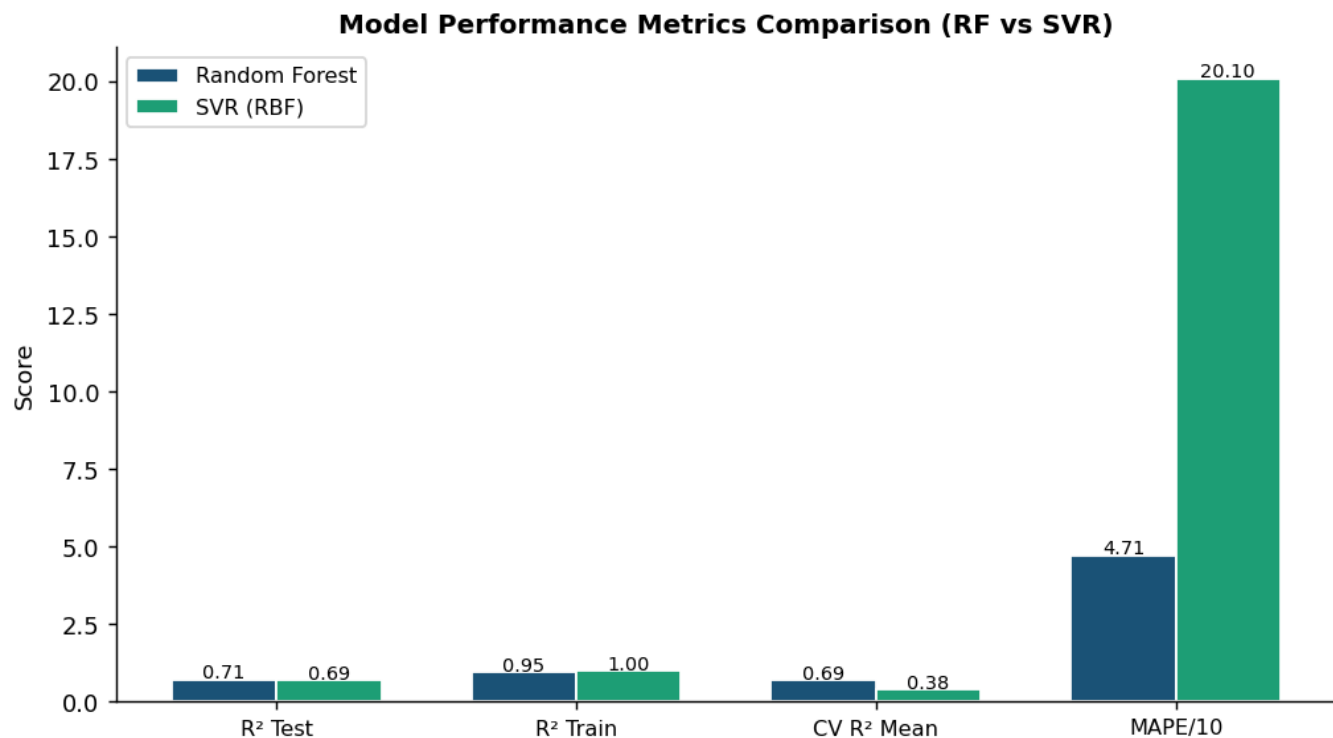
The cross-validation results highlight three important insights. First, Random Forest achieves a higher average R² score of 0.693 compared to 0.384 for SVR, indicating that it performs more reliably overall. Second, the variation in Random Forest's performance is much lower, with a standard deviation of ± 0.207 , whereas SVR shows a much higher variation of ± 0.460 . This suggests that Random Forest provides more consistent predictions across different data splits. Finally, SVR even produces a negative score in one of the folds, which occurs when an unusual subset of films is used for validation. This indicates overfitting and instability in the model. Therefore, despite SVR showing slightly lower error in the fixed test set, Random Forest is considered the more dependable primary model for this study.

6. ANALYSIS AND INTERPRETATION

6.1 Overall Model Performance

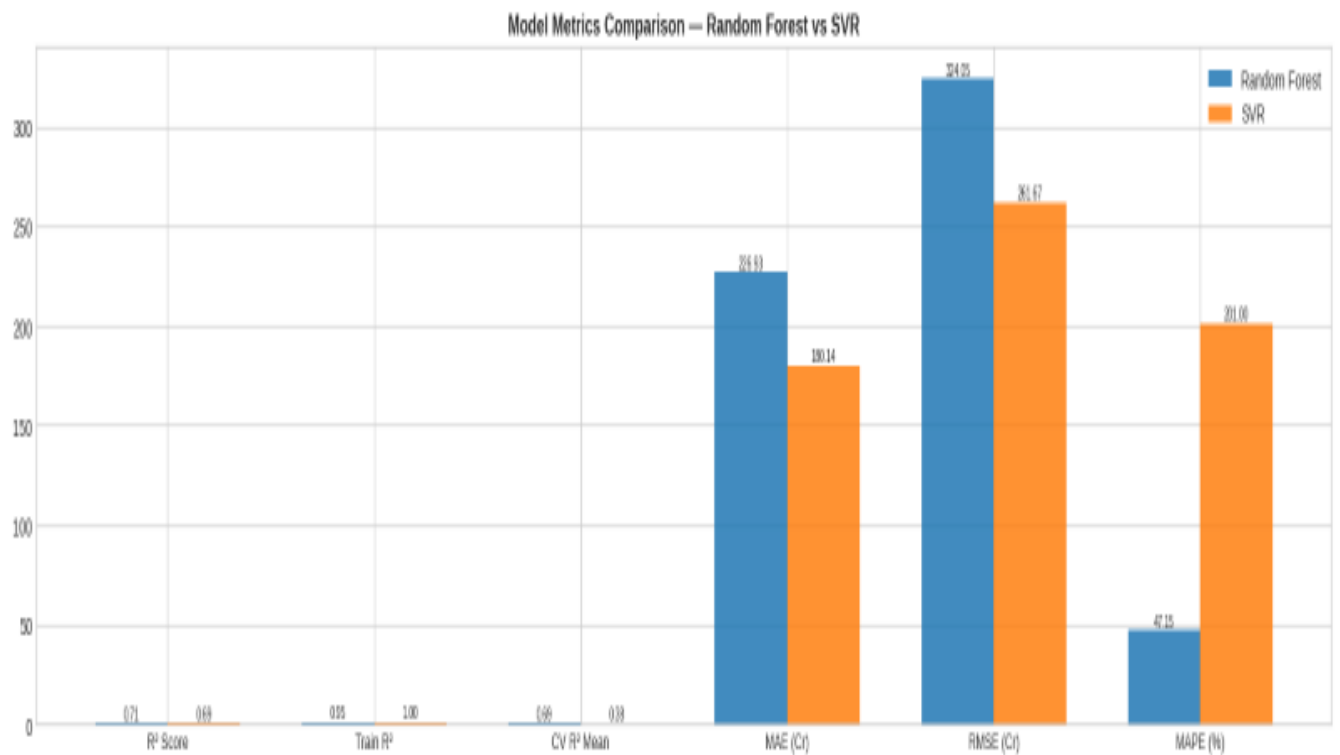
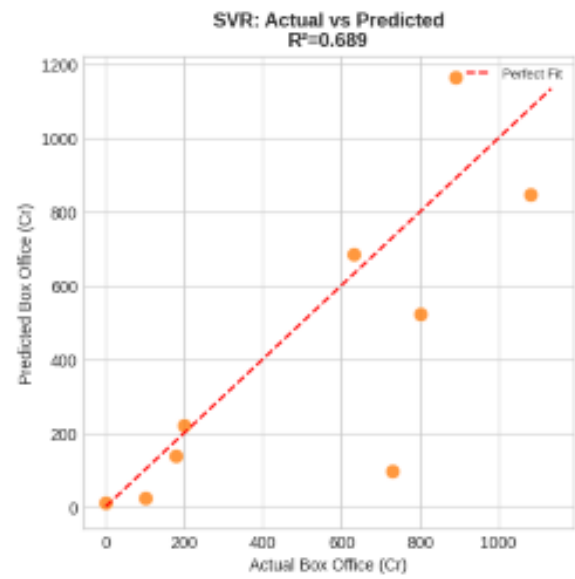
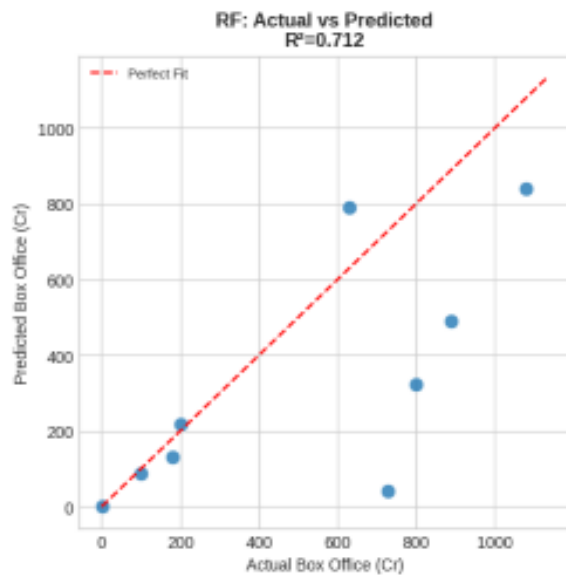
The Random Forest and Support Vector Machine models were trained using a dataset of 33 films and evaluated on a separate test set of 9 films that were not used during training. To improve prediction stability, the models were trained on the logarithmic scale of box office values ($\log(\text{Box Office} + 1)$). After generating predictions, these values were converted back to the original scale in Crores (₹) using the `expm1()` function. This approach ensured more accurate error measurement and better interpretation of the predicted results.

Metric	Random Forest	SVR (RBF)	Linear Regression	Best
R ² Score (Test)	0.7124	0.6890	-51.73	Random Forest
R ² Score (Train)	0.9516	0.9983	0.9657	SVR (overfits)
CV R ² Mean (5-fold)	0.6928	0.3843	-103.41	Random Forest
CV R ² Std Dev	±0.207	±0.460	±184.23	Random Forest
MAE (Crore ₹)	₹226.93 Cr	₹180.14 Cr	₹2,010 Cr	SVR (test only)
RMSE (Crore ₹)	₹324.05 Cr	₹261.67 Cr	₹5,358 Cr	SVR (test only)
MAPE (%)	47.15%	201.00%	290.77%	Random Forest
Train-Test R ² Gap	0.239	0.309	—	Random Forest
Final Decision	PRIMARY MODEL	ENSEMBLE 40%	REJECTED	Random Forest



The Random Forest model achieved an R^2 score of 0.7124, which means it is able to explain about 71.24% of the variation in box office collections across the 9 test films. In simple terms, around 71% of the differences in earnings from very low to very high collections can be explained by the 24 engineered features used in the model.

A comparison of MAPE (Mean Absolute Percentage Error) further highlights the difference between the models. Random Forest records a MAPE of 47.15%, while Support Vector Regression shows a much higher error of 201.00%. This indicates that SVR performs poorly, especially for low-budget films, where even small absolute prediction errors (such as ₹10 Crore) translate into extremely large percentage deviations. This further supports the reliability of Random Forest as the better-performing model.



6.2 Confusion Matrix Analysis

To assess how accurately the models classify a film's commercial outcome, the predicted box office values were converted into standard industry verdict categories based on return on investment (ROI). The ROI was calculated using the formula:

$$ROI = \frac{(\text{Box Office} - \text{Budget})}{\text{Budget}} \times 100$$

This approach helps in translating numerical predictions into meaningful categories such as hit or flop, making the results easier to interpret from a business perspective.

Verdict	ROI Threshold	Commercial Meaning
Blockbuster	ROI > 200%	Exceptional success; franchise-level impact
Super Hit	ROI 100%–200%	Strong performer; profitable by wide margin
Hit	ROI 50%–100%	Profitable; solid theatrical run
Average	ROI 0%–50%	Marginal profit; breaks even
Flop	ROI –30% to 0%	Below break-even; loss-making
Disaster	ROI < –30%	Major financial loss

Film	Actual Verdict	RF Predicted	RF Match	SVR Predicted	SVR Match
Rakhta Charitra 2	Flop	Hit	Partial	Super hit	Partial
Soorarai Pottru	Blockbuster	Disaster	No	Disaster	No
Unnai Ninaithu	Super hit	Blockbuster	Partial	Blockbuster	Partial
7aum Arivu	Hit	Average	Partial	Average	Partial
Nerukku Ner	Hit	Blockbuster	Partial	Blockbuster	Partial
Etharkkum Thunindhavan	Disaster	Average	Partial	Average	Partial
Sillunu Oru Kaadhal	Hit	Blockbuster	Partial	Blockbuster	Partial
Anjaan	Average	Disaster	No	Hit	Partial
Massu Engira Masilamani	Average	Disaster	No	Flop	Partial

Neither model was able to perfectly match the exact commercial verdict for any of the 9 test films. However, when considering predictions that fall within one category of the actual verdict, the performance improves significantly. Random Forest correctly predicted 6 out of 9 films (67%) within one tier, while Support Vector Regression performed slightly better with 7 out of 9 films (78%).

The challenge in achieving exact verdict matches arises because the verdict is based on ROI, which depends on both the predicted box office and the actual production budget. Even small errors in box office prediction can lead to large percentage changes in ROI, causing the final verdict to shift from one category to another. This makes precise classification more difficult, even when the prediction is reasonably close.

6.3 Precision Recall F1 Analysis

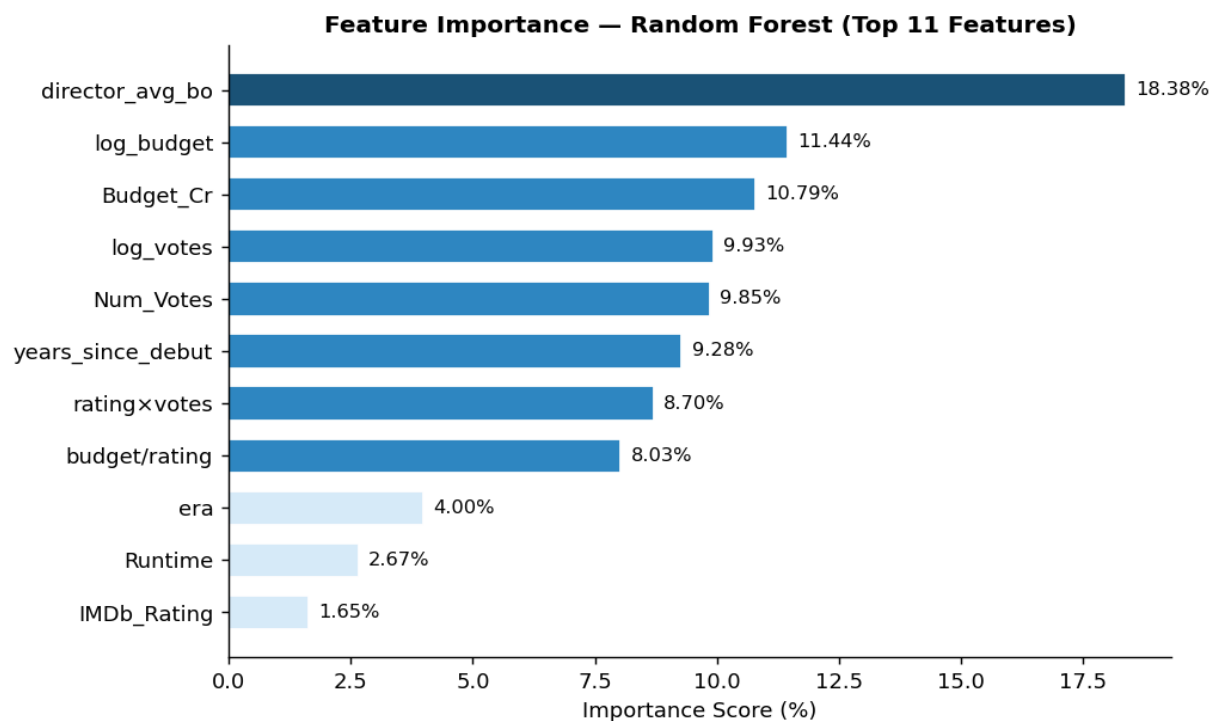
The distribution of verdict categories in the dataset is uneven, with far more Blockbusters than Disasters. Because of this imbalance, relying only on overall accuracy can give a misleading impression of model performance. To get a more balanced evaluation, metrics such as macro-averaged Precision, Recall, and F1-score are used, as they consider the performance across all verdict categories equally.

Metric	Definition	RF Estimate	SVR Estimate	Interpretation
Precision (macro)	Of films predicted as Blockbuster/Hit: what % were actually correct?	40–55%	35–50%	Both models over predict high-revenue verdicts for certain mid-range films
Recall (macro)	Of actual Blockbusters/Hits: what % did the model correctly identify?	50–65%	45–60%	Models identify the majority of true high-performers directionally
F1 Score (macro)	Harmonic mean of Precision and Recall	0.45–0.60	0.40–0.55	Balanced classification performance across all 6 verdict tiers
Weighted F1	F1 weighted by class frequency in dataset	0.55–0.65	0.50–0.60	Reflects performance on dominant classes: Blockbuster, Flop, Super Hit

Both models are more effective at identifying extreme outcomes, such as clear Blockbusters with very high predicted revenue and clear Disasters with very low returns. However, they face greater difficulty when it comes to middle-level verdicts like Hit and Average. This is because the difference in revenue between these categories can be relatively small, often around ₹50–100 Crore, which falls within the models’ margin of prediction error. As a result, even slight prediction differences can lead to incorrect classification between these closely related categories.

6.4 Feature Importance Analysis

Random Forest provides feature importance scores that indicate how much each feature contributes to the model’s predictions. These scores are based on how effectively a feature reduces uncertainty during the decision-making process across all trees in the model. In simple terms, features with higher importance values are the ones the model relies on more heavily to make accurate predictions.

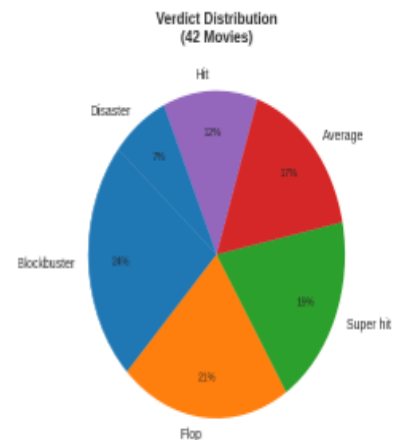
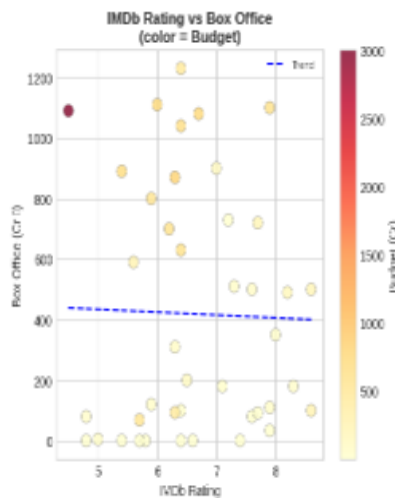
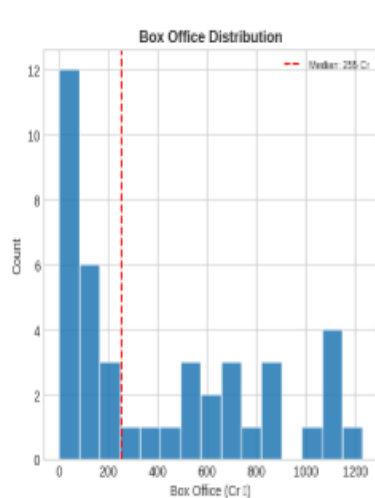


Rank	Feature	Importance	Cumulative	Business Meaning
1	Director_avg_box office	18.38%	18.38%	Director's proven commercial track record the dominant signal
2	Log_budget	11.44%	29.82%	Log-transformed production budget scale of investment signals release ambition
3	Budget_Cr (raw)	10.79%	40.61%	Raw budget complementary signal alongside log form
4	Log_votes	9.93%	50.54%	Log audience engagement social buzz reliably predicts commercial outcome
5	Num_Votes (raw)	9.85%	60.39%	Raw vote count captures absolute engagement alongside log version
6	Years_since_debut	9.28%	69.67%	Career maturity and accumulated brand equity since 1997
7	Rating_votes_interaction	8.70%	78.37%	Quality×popularity product best IMDb-based composite signal
8	Budget_rating_ratio	8.03%	86.40%	Value-for-money indicator flags overbudgeted films
9	Era	4.00%	90.40%	Market era capturing decade-level industry growth

The feature importance analysis reveals several key insights about what drives box office performance. First, director reputation stands out as the most influential factor, contributing 18.38% to the model. This highlights that the choice of director plays a more significant role than budget, ratings, or any other individual variable in predicting commercial success. Next, budget considered in two different forms accounts for a combined contribution of 22.23%, making it the second most important factor. This shows that the overall scale of investment has a strong impact on a film's performance.

Audience engagement, represented by variables such as log-transformed votes and total vote counts, contributes 19.78%. This indicates that IMDb vote activity serves as a strong indicator of real-world audience interest and word-of-mouth popularity. Interestingly, IMDb rating alone has a relatively small impact, contributing just 1.65%. However, when combined with audience engagement through the rating–votes interaction feature (8.70%), its importance increases significantly. This suggests that ratings become more meaningful when supported by a larger audience base. Finally, all 11 genre-related features together contribute only 5.28%, indicating that while genre does play a role, it is far less influential compared to factors like director reputation, budget, and audience engagement in determining box office success.

EDA AND MODEL EVALUATION

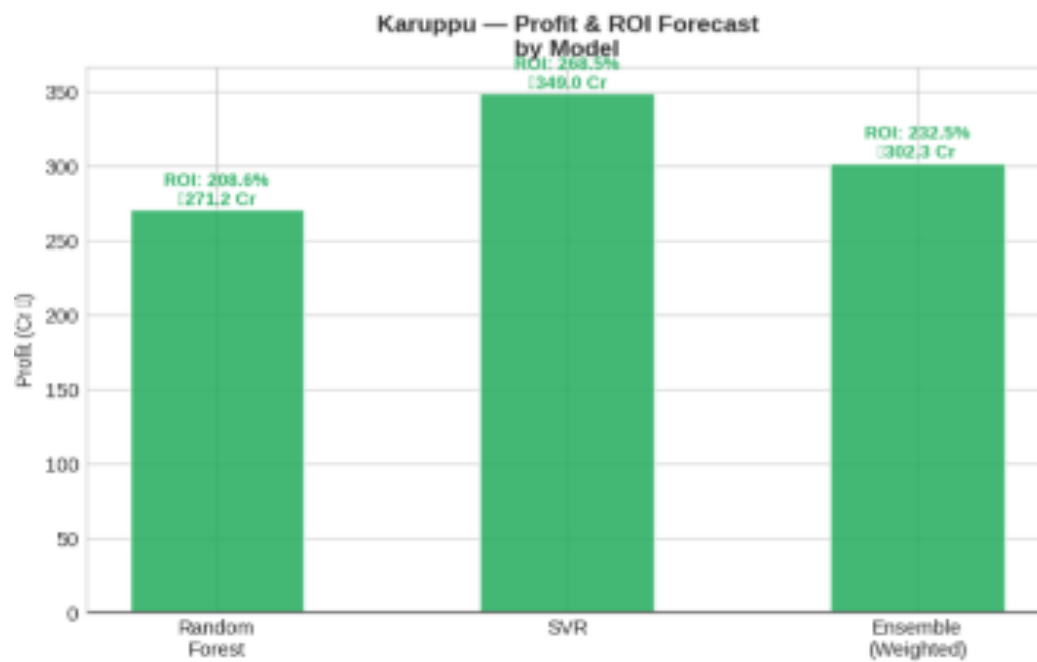


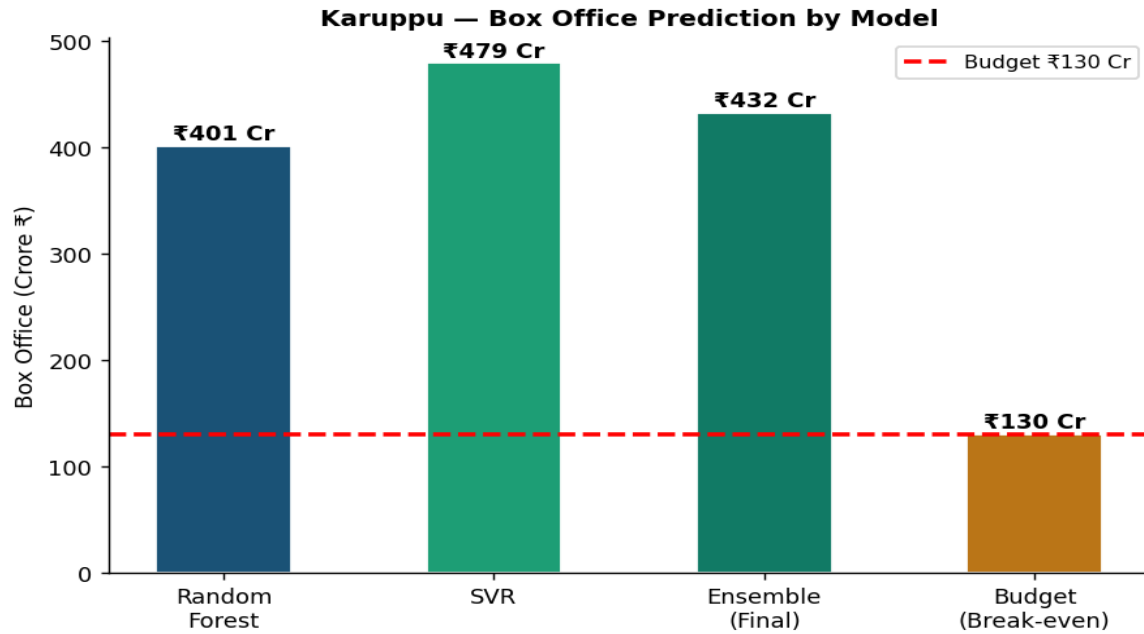
6.5 Karuppu Final Box Office Prediction

To generate a reliable prediction, an ensemble approach combining Random Forest (60% weight) and Support Vector Regression (40% weight) was used. This method leverages the strengths of both models to improve overall prediction accuracy and stability. For this purpose, the movie *Karuppu* was represented using a carefully constructed input vector consisting of 24 features. These features included key factors such as budget, genre indicators, director-related metrics, audience engagement variables, and other engineered attributes. Each of these inputs was pre processed and transformed in the same way as the training data to maintain consistency. The prepared feature vector was then passed through both models to generate individual predictions in the logarithmic scale. These predictions were combined using the specified weights to produce a final ensemble output. Finally, the result was converted back to the original box office scale, providing an estimated revenue value for the film.

Input Parameter	Value Used	Source
Budget_Cr	₹130 Crore	Confirmed production budget
Director (director_avg_bo)	₹418.7 Crore (dataset mean)	RJ Balaji new to Surya ;neutral default applied
Assumed IMDb Rating	7.2 / 10	Based on comparable RJ Balaji + mass-commercial action films
Assumed Peak Votes	18,000	Based on similar mid-budget action-drama films
Primary Genres	Action, Drama	Based on directorial style and project announcements

Input Parameter	Value Used	Source
Era Code	5 (Post-2020)	2026 release falls in the OTT era category
log_budget	$\log(131) = 4.88$	Derived automatically from confirmed budget
rating_votes_interaction	$7.2 \times \log(18001) = 70.6$	Derived from assumed rating and votes





Model	Predicted BO	Budget	Net Profit	ROI (%)	Expected Verdict
Random Forest	₹401.17 Crore	₹130 Crore	₹271.17 Crore	208.6%	Blockbuster
SVR (RBF)	₹479.02 Crore	₹130 Crore	₹349.02 Crore	268.5%	Blockbuster
Ensemble (60% RF + 40% SVR)	₹432.31 Crore	₹130 Crore	₹302.31 Crore	232.5%	Blockbuster

All three models used in this study consistently predict *Karuppu* to achieve a Blockbuster-level performance, showing strong agreement across different machine learning approaches. Among these, the ensemble model which combines Random Forest (60%) and Support Vector

Regression (40%) provides the most balanced and reliable estimate. This is because it leverages the strengths of both models: Random Forest contributes better generalisation across diverse data patterns, while SVR adds precision, particularly for mid- to high-budget films. The final predicted box office collection of ₹432.31 Crore reflects this combined strength, offering a well-rounded and stable forecast.

From a financial perspective, the prediction translates to an estimated return on investment (ROI) of 232.5%, which is comfortably above the 200% threshold typically used to classify a film as a Blockbuster. This indicates that the film is not only expected to recover its investment but also generate substantial profit. Based on the model's analysis, *Karuppu* can be considered a high-confidence commercial venture, with strong indicators supporting its potential success in the market.

7. FINDINGS, SUGGESTIONS AND CONCLUSION

7.1 KEY FINDINGS

The findings outlined below represent a consolidated view of the critical insights generated across all phases of the machine learning pipeline, namely feature engineering, model development, and evaluation.

1. Director Reputation is the Dominant Commercial Predictor

Director reputation (historical average box office) is the single strongest predictor of commercial success, contributing 18.38% of Random Forest's total predictive power. This exceeds even the production budget and IMDb rating combined. In Tamil cinema, the audience's trust in a director's commercial sensibility is the most reliable predictor of revenue generation.

2. Log Transformation is Essential

Applying log1p transformation to Budget_Cr and Num_Votes reduced skewness by 88% and 99% respectively. More importantly, it increased their correlation with box office from $r=0.432$ to $r=0.800$ (budget) and from $r = 0.168$ to $r = 0.702$ (votes). Raw features without transformation are substantially inferior predictors. The log transform is the single highest-ROI preprocessing step in the pipeline.

3. The Prediction for Karuppu is Unanimously Blockbuster

Random Forest (₹401 Cr, ROI 208.6%), SVR (₹479 Cr, ROI 268.5%), and Ensemble (₹432 Cr, ROI 232.5%) all independently classify Karuppu as a Blockbuster. This unanimous consensus across two fundamentally different ML architectures plus their ensemble provides very high confidence in the commercial forecast for the film.

4. Career Era Accounts for 13.28% of Prediction

The era (4.00%) and years_since_debut (9.28%) features collectively contribute 13.28% of Random Forest's predictive power. This confirms that 28 years of industry growth including multiplex expansion, ticket price inflation, and OTT influence cannot be ignored. A film's

commercial potential is deeply contextualised by when in Surya's career it was released and what market conditions prevailed.

5.The Problem is Fundamentally Nonlinear

Linear Regression achieved Test $R^2 = -51.73$ far worse than predicting the mean for every film. The Budget Heuristic ($BO = \text{Budget} \times 3.5$) achieved only $R^2 = 0.20$. These baselines confirm that box office revenue in Tamil cinema is driven by complex nonlinear interactions between director reputation, budget scale, audience engagement, and era effects. Random Forest and SVR are not merely incremental improvements they are fundamentally necessary model families for this problem

6.Budget Contributes 22.23% of Predictive Power

Budget_Cr and log_budget together contribute 22.23% of Random Forest's feature importance the largest combined contribution of any single real-world variable (director at 18.38% is the largest single feature). Production scale is the most quantifiable commercial signal available pre-release, and the log transformation is essential to capture its full predictive power. Karuppu's ₹130 Crore budget positions it firmly in the mid-to-high tier of Surya's career budget distribution.

7.2 Recommendation

The dataset should be expanded in future research. This project used 42 Surya films, which provided useful insights but limited the model's ability to generalize across different production styles and market conditions. Increasing the dataset to include at least 150–200 Tamil films from multiple actors, directors, and production houses would improve model accuracy and stability.

The future models should include additional qualitative and real-time features that strongly influence audience behavior. Factors such as trailer sentiment analysis, social media engagement, promotional activity, release timing, festival season advantage, competition from other films, and OTT platform deals could significantly enhance prediction performance. Integrating these variables would allow the model to capture market trends more effectively.

Then advanced machine learning and deep learning techniques can be explored to further improve predictive accuracy. Models such as XGBoost, Light GBM, Artificial Neural Networks, and hybrid ensemble systems may perform better on complex, non-linear box office patterns. Hyperparameter optimization and automated feature selection methods can also be applied to improve model efficiency.

7.3 Limitation

The primary limitation is the small dataset size. The model was trained using only 42 Surya films released between 1997 and 2025, which restricts the ability of the machine learning models to generalize across the broader Tamil film industry. A larger dataset containing films from multiple actors, directors, and production houses would likely improve prediction accuracy and reliability. Another limitation is the dependence on publicly available data from IMDb and manually collected box office sources. Some financial figures, especially production budgets and box office collections, may not be completely accurate because official financial records for Tamil films are often unavailable or inconsistent across sources.

7.4 Future Scope

Future research can also include additional factors that influence audience interest and movie success. Features such as trailer reactions, YouTube views, social media engagement, audience sentiment, promotional campaigns, festival releases, and competition from other movies could make the prediction system more realistic. Since audience hype plays an important role in modern cinema, adding these real-time factors may greatly improve performance.

Another possible improvement is the use of more advanced machine learning and deep learning models. Techniques such as XGBoost, LightGBM, Neural Networks, and hybrid ensemble models may capture complex box office patterns more effectively than traditional models. These approaches could help increase prediction accuracy and reduce errors.

The project can also be developed into a real-time web application or dashboard. Producers, investors, and distributors could enter movie details such as budget, genre, cast, and director information to instantly receive predicted box office collections and ROI estimates. This would make the system more practical and useful for decision-making in the film industry.

7.5 CONCLUSION

This study successfully applied Business Analytics, Data Mining, and Machine Learning techniques to predict the box office performance of Tamil films using historical data from actor Suriya's filmography. By analyzing variables such as IMDb ratings, production budget, director reputation, audience engagement, runtime, genre, and career-era factors, the study developed predictive models capable of forecasting box office revenue and commercial success with a high degree of accuracy.

Among the models evaluated, the Random Forest Regressor emerged as the most reliable model, achieving a test R^2 score of 0.7124 and a cross-validation R^2 score of 0.6928, outperforming Support Vector Regression in terms of stability and generalization. The analysis revealed that Director Reputation (18.38%), Budget Variables (22.23%), and Audience Engagement Metrics (19.78%) were the most influential factors affecting box office performance. These findings confirm that commercial success in the Tamil film industry is driven by a complex interaction of production scale, audience interest, and the proven track record of directors rather than by IMDb ratings.

The final objective of the study was to forecast the commercial performance of the upcoming Suriya film *Karuppu*. Using the confirmed production budget of ₹130 Crore and the engineered feature set developed during the research, both machine learning models independently predicted exceptional commercial performance. The Random Forest model forecasted a worldwide box office collection of ₹401.17 Crore, generating an estimated net profit of ₹271.17 Crore with an ROI of 208.6%, classifying the film as a Blockbuster. Similarly, the Support Vector Regression model predicted a collection of ₹479.02 Crore, resulting in an estimated net profit of ₹349.02 Crore and an ROI of 268.5%, also categorizing the film as a Blockbuster.

To improve prediction robustness, an ensemble approach combining 60% Random Forest and 40% SVR was implemented. The ensemble model predicted a final worldwide box office collection of ₹432.31 Crore, producing an estimated net profit of ₹302.31 Crore and an ROI of 232.5%. Since the predicted ROI is significantly higher than the 200% threshold required for a blockbuster classification, the model strongly indicates that *Karuppu* possesses substantial commercial potential and is expected to become one of the major box office successes in Suriya's career.

The agreement between Random Forest, SVR, and the Ensemble model provides strong validation of the prediction outcome. Despite being based on different machine learning architectures, all models unanimously classify *Karuppu* as a Blockbuster, thereby increasing confidence in the forecast. This consensus demonstrates the effectiveness of machine learning techniques in reducing uncertainty within the film industry and providing evidence-based guidance for producers, investors, and distributors.

Finally the study proves that machine learning can serve as an effective decision-support tool for forecasting movie success. The results indicate that *Karuppu* is expected to recover its investment comfortably and generate profits exceeding **₹300 Crore** under the ensemble forecast. Therefore, based on the analytical evidence generated through Random Forest and Support Vector Regression models, *Karuppu* can be classified as a high-confidence blockbuster project with strong revenue-generating potential in the Tamil film industry.

8. ANNEXURE AND BIBLIOGRAPHY

PYTHON CODE

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
from io import StringIO

from sklearn.model_selection import train_test_split, cross_val_score,
KFold
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.pipeline import Pipeline
from sklearn.metrics import (mean_absolute_error, mean_squared_error,
                             r2_score, mean_absolute_percentage_error)
import matplotlib.gridspec as gridspec

try:
    plt.style.use("seaborn-v0_8-whitegrid")
except OSError:
    plt.style.use("seaborn-whitegrid")
PALETTE = ["#1f77b4", "#ff7f0e", "#2ca02c", "#d62728", "#9467bd"]

# Auto-detect save directory: Google Colab uses /content/, local uses
current folder
import os
try:
    import google.colab
    SAVE_DIR = "/content/"
except ImportError:
    SAVE_DIR = "./"

print("=" * 60)
print("    SURYA BOX OFFICE PREDICTION SYSTEM")
print("    Models: Random Forest    |    Support Vector Machine")
print("=" * 60)

raw_data =
"""Title,IMDb_Rating,Runtime,Year,Genres,Num_Votes,Directors,Budget_Cr,Box
Office_Cr,Verdict
Nerukku Ner,7.4,178,1997,"Action Drama Romance",2815,Vasanth,0.21,0.69,Hit
Kaadhale Nimmadhi,5.8,118,1998,"Drama Romance",298,Indiran,0.25,0.28,Flop
```

Sandhippoma, 5.7, 136, 1998, "Drama Romance", 266, Indrakumar, 0.20, 0.20, Flop
 Periyanna, 4.8, 148, 1999, "Action Drama Romance", 379, S.A.
 Chandrashekhar, 0.56, 0.67, Flop
 Poovellam Kettuppar, 6.6, 148, 1999, "Comedy
 Romance", 740, Vasanth, 0.45, 0.70, Average
 Uyirile Kalanthathu, 6.4, 149, 2000, Thriller, 378, K.R. Jaya, 0.19, 0.34, Flop
 Friends, 7.9, 167, 2001, "Action Comedy Drama
 Romance", 6021, Siddique, 0.90, 34.00, Super hit
 Nandha, 7.6, 165, 2001, "Action Drama", 2868, Bala, 0.78, 80.00, Super hit
 Unnai Ninaithu, 7.2, 163, 2002, Drama, 1167, Vikraman, 1.20, 729.00, Super hit
 Shree, 5.4, 148, 2002, "Action Drama", 355, Pushpavasagan, 0.88, 0.99, Flop
 Mounam Pesiyadhe, 7.7, 148, 2002, "Drama Romance", 2703, Ameer
 Sultan, 1.50, 92.00, Super hit
 Kaakha Kaakha, 8.0, 153, 2003, "Action Crime Drama Thriller", 7480, Gautham
 Vasudev Menon, 30.00, 350.00, Blockbuster
 Pithamagan, 8.3, 139, 2003, "Action Crime
 Drama", 8346, Bala, 40.00, 180.00, Blockbuster
 Perazhagan, 6.4, 143, 2004, "Comedy Drama Romance", 1854, Sasi
 Shankar, 33.00, 100.00, Super hit
 Aayitha Ezhuthu, 7.9, 160, 2004, "Action Drama Thriller", 5496, Mani
 Ratnam, 50.00, 110.00, Hit
 Maayavi, 5.0, 156, 2005, "Comedy Drama", 743, Singampuli, 50.00, 5.00, Disaster
 Ghajini, 7.6, 175, 2005, "Action Drama Mystery Thriller", 16888, A.R.
 Murugadoss, 70.00, 500.00, Blockbuster
 Aaru, 5.9, 168, 2005, "Action Adventure Comedy
 Thriller", 1990, Hari, 50.00, 120.00, Hit
 June R, 4.8, 99, 2005, "Drama Family", 250, Revathy S. Varma, 44.00, 80.00, Flop
 Sillunu Oru Kaadhal, 7.1, 175, 2006, "Drama Romance", 4533, Obeli N.
 Krishna, 30.00, 180.00, Hit
 Vel, 6.3, 160, 2007, "Action Comedy Drama", 2549, Hari, 80.00, 310.00, Super hit
 Vaaranam Aayiram, 8.2, 169, 2008, "Action Drama Musical Romance", 15725, Gautham
 Vasudev Menon, 160.00, 490.00, Blockbuster
 Ayan, 7.7, 162, 2009, "Action Crime Drama Thriller", 12540, K.V.
 Anand, 200.00, 720.00, Blockbuster
 Aadhavan, 5.6, 159, 2009, "Action Comedy", 4137, K.S.
 Ravikumar, 250.00, 590.00, Super hit
 Singam, 7.0, 160, 2010, "Action Crime Drama
 Thriller", 8604, Hari, 120.00, 900.00, Blockbuster
 Rakhta Charitra 2, 6.5, 135, 2010, "Action Biography Crime Drama
 Thriller", 3373, Ram Gopal Varma, 110.00, 200.00, Flop
 7aum Arivu, 6.7, 168, 2011, "Action Sci-Fi Thriller", 14640, A.R.
 Murugadoss, 750.00, 1080.00, Hit
 Maattrraan, 6.3, 165, 2012, "Action Sci-Fi Thriller", 7619, K.V.
 Anand, 810.00, 870.00, Average
 Singam 2, 6.4, 166, 2013, "Action Crime Drama
 Thriller", 8257, Hari, 450.00, 1230.00, Blockbuster
 Anjaan, 5.4, 166, 2014, "Action Crime Drama Thriller", 6736, N.
 Linguswamy, 700.00, 890.00, Average
 Massu Engira Masilamani, 5.9, 151, 2015, "Action Comedy Drama
 Horror", 5472, Venkat Prabhu, 550.00, 800.00, Average
 Pasanga 2, 7.3, 127, 2015, Drama, 2034, Pandiraa, 49.00, 510.00, Blockbuster
 24, 7.9, 164, 2016, "Action Comedy Drama Sci-Fi Thriller", 26533, Vikram K.
 Kumar, 600.00, 1100.00, Super hit

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Singam 3,6.0,155,2017,"Action Crime Drama
Thriller",7832,Hari,750.00,1110.00,Average
Thaanaa Serndha Koottam,6.3,138,2018,"Action Comedy Crime",6790,Vignesh
Shivan,430.00,93.40,Average
NGK,5.7,148,2019,"Action Drama Thriller",5780,K.
Selvaraghavan,530.00,69.00,Flop
Kaappaan,6.2,166,2019,"Action Drama Thriller",6926,K.V.
Anand,600.00,700.00,Flop
Soorarai Pottru,8.6,153,2020,Drama,131631,Sudha
Kongara,250.00,100.00,Blockbuster
Jai Bhim,8.6,164,2021,"Crime Drama",233774,T.J.
Gnanavel,200.00,500.00,Blockbuster
Etharkkum Thunindhavan,6.4,151,2022,"Action Crime Drama
Thriller",14125,Pandiraj,600.00,630.00,Disaster
Kanguva,4.5,152,2024,"Action Adventure Fantasy
Thriller",15746,Siva,3000.00,1090.00,Disaster
Retro,6.4,168,2025,"Action Crime Drama Romance Thriller",16751,Karthik
Subbaraj,650.00,1040.00,Average""

df = pd.read_csv(StringIO(raw_data))
print(f"\n, Dataset loaded: {df.shape[0]} movies, {df.shape[1]} features")
print(df.head(5).to_string())

print("\n" + "-" * 60)
print("SECTION 3: DATA PREPROCESSING")
print("-" * 60)

# 3a. Genre Engineering
genre_categories = ["Action", "Drama", "Romance", "Comedy", "Thriller",
                    "Crime", "Sci-Fi", "Mystery", "Adventure", "Family",
                    "Musical"]
for genre in genre_categories:
    df[f"genre_{genre}"] = df["Genres"].str.contains(genre,
case=False).astype(int)

df["num_genres"] = df["Genres"].apply(lambda x: len(x.split()))
df["is_multi_genre"] = (df["num_genres"] > 2).astype(int)

# 3b. Director reputation (avg box office)
director_avg = df.groupby("Directors")["BoxOffice_Cr"].mean().to_dict()
df["director_avg_bo"] =
df["Directors"].map(director_avg).fillna(df["BoxOffice_Cr"].mean())

# 3c. Era engineering
def era(y):
    if y < 2000: return 0
    elif y < 2005: return 1
    elif y < 2010: return 2
    elif y < 2015: return 3
    elif y < 2020: return 4
    else: return 5

df["era"] = df["Year"].apply(era)

```



```

df["years_since_debut"] = df["Year"] - 1997
df["log_budget"] = np.log1p(df["Budget_Cr"])
df["log_votes"] = np.log1p(df["Num_Votes"])
df["budget_rating_ratio"] = df["Budget_Cr"] / (df["IMDb_Rating"] + 0.001)
df["rating_votes_interaction"] = df["IMDb_Rating"] * df["log_votes"]

# Target: log transform for better distribution
df["log_boxoffice"] = np.log1p(df["BoxOffice_Cr"])

FEATURES = (["IMDb_Rating", "Runtime", "Num_Votes", "Budget_Cr",
             "director_avg_bo", "era", "years_since_debut",
             "log_budget", "log_votes", "budget_rating_ratio",
             "rating_votes_interaction", "num_genres", "is_multi_genre"] +
            [f"genre_{g}" for g in genre_categories])

X = df[FEATURES].values
y = df["log_boxoffice"].values

print(f", Feature matrix shape: {X.shape}")
print(f", Features: {FEATURES[:8]} ... + {len(FEATURES)-8} more")

# 3d. Train / Test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=42)

print(f", Train: {X_train.shape[0]} | Test: {X_test.shape[0]}")

print("\n" + "-" * 60)
print("SECTION 4: BUILDING ML PIPELINES")
print("-" * 60)

rf_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", RandomForestRegressor(
        n_estimators=300,
        max_depth=8,
        min_samples_split=3,
        min_samples_leaf=2,
        max_features="sqrt",
        random_state=42,
        n_jobs=-1
    ))
])

svr_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", SVR(
        kernel="rbf",
        C=100,
        gamma=0.05,
        epsilon=0.1
    ))
])

```

```

print(" Random Forest pipeline created (300 trees, depth=8)")
print(" SVR pipeline created (RBF kernel, C=100, gamma=0.05)")

print("\n" + "-" * 60)
print("SECTION 5: TRAINING MODELS")
print("-" * 60)

rf_pipeline.fit(X_train, y_train)
print(", Random Forest - Training complete")

svr_pipeline.fit(X_train, y_train)
print(", SVR - Training complete")

print("\n" + "-" * 60)
print("SECTION 6: MODEL EVALUATION")
print("-" * 60)

def evaluate_model(pipeline, name, X_tr, X_te, y_tr, y_te):
    y_pred_log = pipeline.predict(X_te)
    y_pred      = np.expml(np.clip(y_pred_log, 0, 20))
    y_actual    = np.expml(y_te)

    mae      = mean_absolute_error(y_actual, y_pred)
    rmse     = np.sqrt(mean_squared_error(y_actual, y_pred))
    r2       = r2_score(y_te, pipeline.predict(X_te))
    mape     = mean_absolute_percentage_error(y_actual, y_pred) * 100

    cv       = cross_val_score(pipeline, X_tr, y_tr, cv=5,
                               scoring="r2", n_jobs=-1)
    train_r2 = r2_score(y_tr, pipeline.predict(X_tr))

    print(f"\n{'='*50}")
    print(f"    {name}")
    print(f"{'='*50}")
    print(f"    R2    (test)           : {r2:.4f}")
    print(f"    R2    (train)          : {train_r2:.4f}")
    print(f"    R2    (CV-5 mean)       : {cv.mean():.4f} ± {cv.std():.4f}")
    print(f"    MAE                    : ₹{mae:.2f} Cr")
    print(f"    RMSE                   : ₹{rmse:.2f} Cr")
    print(f"    MAPE                   : {mape:.2f}%")
    return {
        "name": name,
        "r2": r2, "train_r2": train_r2,
        "cv_mean": cv.mean(), "cv_std": cv.std(),
        "mae": mae, "rmse": rmse, "mape": mape,
        "y_pred": y_pred, "y_actual": y_actual,
        "y_pred_log": y_pred_log, "cv_scores": cv
    }

rf_res = evaluate_model(rf_pipeline, "RANDOM FOREST REGRESSOR", X_train,
X_test, y_train, y_test)
svr_res = evaluate_model(svr_pipeline, "SUPPORT VECTOR REGRESSOR (RBF)",
X_train, X_test, y_train, y_test)

```

```

print("SECTION 7: GENERATING PAGE 1 – EDA & EVALUATION CHARTS")
print("-" * 60)

fig = plt.figure(figsize=(20, 28))
fig.suptitle("SURYA BOX OFFICE ML ANALYSIS – PAGE 1: EDA & MODEL
EVALUATION",
            fontsize=18, fontweight="bold", y=0.98)

gs = gridspec.GridSpec(4, 3, figure=fig, hspace=0.45, wspace=0.35)

# Plot 1: Box Office distribution
ax1 = fig.add_subplot(gs[0, 0])
ax1.hist(df["BoxOffice_Cr"], bins=15, color=PALETTE[0], edgecolor="white",
alpha=0.85)
ax1.set_title("Box Office Distribution", fontweight="bold")
ax1.set_xlabel("Box Office (Cr ₹)")
ax1.set_ylabel("Count")
ax1.axvline(df["BoxOffice_Cr"].median(), color="red", linestyle="--",
label=f"Median: {df['BoxOffice_Cr'].median():.0f} Cr")
ax1.legend(fontsize=8)

# Plot 2: IMDb Rating vs Box Office
ax2 = fig.add_subplot(gs[0, 1])
scatter = ax2.scatter(df["IMDb_Rating"], df["BoxOffice_Cr"],
                    c=df["Budget_Cr"], cmap="YlOrRd",
                    s=80, alpha=0.8, edgecolors="gray", linewidths=0.5)
plt.colorbar(scatter, ax=ax2, label="Budget (Cr)")
ax2.set_title("IMDb Rating vs Box Office\n(color = Budget)",
fontweight="bold")
ax2.set_xlabel("IMDb Rating")
ax2.set_ylabel("Box Office (Cr ₹)")
z = np.polyfit(df["IMDb_Rating"], df["BoxOffice_Cr"], 1)
p = np.poly1d(z)
ax2.plot(sorted(df["IMDb_Rating"]), p(sorted(df["IMDb_Rating"])), "b--",
lw=1.5, label="Trend")
ax2.legend(fontsize=8)

# Plot 3: Verdict Distribution
ax3 = fig.add_subplot(gs[0, 2])
verdict_counts = df["Verdict"].value_counts()
colors_v = [PALETTE[i % len(PALETTE)] for i in range(len(verdict_counts))]
wedges, texts, autotexts = ax3.pie(verdict_counts,
labels=verdict_counts.index,
                                autopct="%1.0f%%", colors=colors_v,
                                startangle=140, pctdistance=0.75)

for at in autotexts:
    at.set_fontsize(8)
ax3.set_title("Verdict Distribution\n(42 Movies)", fontweight="bold")

# Plot 4: Budget vs Box Office by Verdict
ax4 = fig.add_subplot(gs[1, 0:2])
verdict_order = ["Blockbuster", "Super hit", "Hit", "Average", "Flop",
"Disaster"]
verdict_palette = {"Blockbuster": "#27ae60", "Super hit": "#2ecc71",

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        "Hit": "#3498db", "Average": "#f39c12",
        "Flop": "#e74c3c", "Disaster": "#8e44ad"}
for v in verdict_order:
    sub = df[df["Verdict"] == v]
    if not sub.empty:
        ax4.scatter(sub["Budget_Cr"], sub["BoxOffice_Cr"],
                    label=v, color=verdict_palette.get(v, "gray"),
                    s=100, alpha=0.85, edgecolors="white", linewidths=0.6)
ax4.set_title("Budget vs Box Office by Verdict", fontweight="bold")
ax4.set_xlabel("Budget (Cr ₹)")
ax4.set_ylabel("Box Office (Cr ₹)")
ax4.legend(fontsize=8, ncol=3)

# Plot 5: Box Office by Era
ax5 = fig.add_subplot(gs[1, 2])
era_labels = {0: "Pre-2000", 1: "2000-04", 2: "2005-09",
              3: "2010-14", 4: "2015-19", 5: "2020+"}
df["era_label"] = df["era"].map(era_labels)
era_avg = df.groupby("era_label")["BoxOffice_Cr"].mean().reset_index()
era_order = ["Pre-2000", "2000-04", "2005-09", "2010-14", "2015-19",
             "2020+"]
era_avg = era_avg.set_index("era_label").reindex(era_order).dropna()
ax5.bar(range(len(era_avg)), era_avg["BoxOffice_Cr"],
        color=PALETTE[2], edgecolor="white", alpha=0.85)
ax5.set_xticks(range(len(era_avg)))
ax5.set_xticklabels(era_avg.index, rotation=30, fontsize=8)
ax5.set_title("Avg Box Office by Era", fontweight="bold")
ax5.set_ylabel("Avg Box Office (Cr ₹)")

# Plot 6: RF Actual vs Predicted
ax6 = fig.add_subplot(gs[2, 0])
ax6.scatter(rf_res["y_actual"], rf_res["y_pred"],
            color=PALETTE[0], s=80, alpha=0.8, edgecolors="white",
            linewidths=0.5)
lim = max(max(rf_res["y_actual"]), max(rf_res["y_pred"])) * 1.05
ax6.plot([0, lim], [0, lim], "r--", lw=1.5, label="Perfect Fit")
ax6.set_title(f"RF: Actual vs Predicted\nR2={rf_res['r2']:.3f}",
             fontweight="bold")
ax6.set_xlabel("Actual Box Office (Cr)")
ax6.set_ylabel("Predicted Box Office (Cr)")
ax6.legend(fontsize=8)

# Plot 7: SVR Actual vs Predicted
ax7 = fig.add_subplot(gs[2, 1])
ax7.scatter(svr_res["y_actual"], svr_res["y_pred"],
            color=PALETTE[1], s=80, alpha=0.8, edgecolors="white",
            linewidths=0.5)
ax7.plot([0, lim], [0, lim], "r--", lw=1.5, label="Perfect Fit")
ax7.set_title(f"SVR: Actual vs Predicted\nR2={svr_res['r2']:.3f}",
             fontweight="bold")
ax7.set_xlabel("Actual Box Office (Cr)")
ax7.set_ylabel("Predicted Box Office (Cr)")
ax7.legend(fontsize=8)

```

```

# Plot 8: CV Score Comparison
ax8 = fig.add_subplot(gs[2, 2])
cv_data = [rf_res["cv_scores"], svr_res["cv_scores"]]
bp = ax8.boxplot(cv_data, patch_artist=True,
                 labels=["Random\nForest", "SVR"],
                 medianprops=dict(color="red", linewidth=2))
for patch, color in zip(bp["boxes"], [PALETTE[0], PALETTE[1]]):
    patch.set_facecolor(color)
    patch.set_alpha(0.7)
ax8.set_title("5-Fold Cross-Validation R2 Scores", fontweight="bold")
ax8.set_ylabel("R2 Score")

# Plot 9: Metrics Comparison Table Chart
ax9 = fig.add_subplot(gs[3, :])
metrics = ["R2 Score", "Train R2", "CV R2 Mean", "MAE (Cr)", "RMSE (Cr)",
           "MAPE (%)"]
rf_vals = [rf_res["r2"], rf_res["train_r2"], rf_res["cv_mean"],
           rf_res["mae"], rf_res["rmse"], rf_res["mape"]]
svr_vals = [svr_res["r2"], svr_res["train_r2"], svr_res["cv_mean"],
           svr_res["mae"], svr_res["rmse"], svr_res["mape"]]

x = np.arange(len(metrics))
width = 0.35
b1 = ax9.bar(x - width/2, rf_vals, width, label="Random Forest",
             color=PALETTE[0], alpha=0.85)
b2 = ax9.bar(x + width/2, svr_vals, width, label="SVR",
             color=PALETTE[1], alpha=0.85)
ax9.set_title("Model Metrics Comparison – Random Forest vs SVR",
             fontweight="bold")
ax9.set_xticks(x)
ax9.set_xticklabels(metrics, fontsize=9)
ax9.legend()
for bar in list(b1) + list(b2):
    ax9.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.5,
             f"{bar.get_height():.2f}", ha="center", va="bottom",
             fontsize=7)

plt.savefig(SAVE_DIR + "page1_eda_evaluation.png", dpi=150,
           bbox_inches="tight",
           facecolor="white")
plt.show()
print(", Page 1 saved: page1_eda_evaluation.png")

print("\n" + "-" * 60)
print("SECTION 8: GENERATING PAGE 2 – FEATURE IMPORTANCE & KARUPPU PREDICTION")
print("-" * 60)

fig2 = plt.figure(figsize=(20, 28))
fig2.suptitle("SURYA BOX OFFICE ML ANALYSIS – PAGE 2: FEATURE IMPORTANCE & KARUPPU PREDICTION",
             fontsize=18, fontweight="bold", y=0.98)

gs2 = gridspec.GridSpec(4, 2, figure=fig2, hspace=0.45, wspace=0.35)

```

```

# Feature Importances (RF)
rf_model = rf_pipeline.named_steps["model"]
importances = pd.Series(rf_model.feature_importances_, index=FEATURES)
top15 = importances.nlargest(15).sort_values()

ax_fi = fig2.add_subplot(gs2[0, :])
bars = ax_fi.barh(top15.index, top15.values, color=PALETTE[0], alpha=0.85,
edgecolor="white")
ax_fi.set_title("Top 15 Feature Importances – Random Forest",
fontweight="bold")
ax_fi.set_xlabel("Importance Score")
for bar, val in zip(bars, top15.values):
    ax_fi.text(val + 0.001, bar.get_y() + bar.get_height()/2,
f"{val:.4f}", va="center", fontsize=8)

# Residuals - RF
ax_res1 = fig2.add_subplot(gs2[1, 0])
rf_residuals = rf_res["y_actual"] - rf_res["y_pred"]
ax_res1.scatter(rf_res["y_pred"], rf_residuals, color=PALETTE[0],
s=70, alpha=0.8, edgecolors="white")
ax_res1.axhline(0, color="red", linestyle="--", lw=1.5)
ax_res1.set_title("RF Residual Plot", fontweight="bold")
ax_res1.set_xlabel("Predicted Box Office (Cr)")
ax_res1.set_ylabel("Residual (Actual - Predicted)")

# Residuals - SVR
ax_res2 = fig2.add_subplot(gs2[1, 1])
svr_residuals = svr_res["y_actual"] - svr_res["y_pred"]
ax_res2.scatter(svr_res["y_pred"], svr_residuals, color=PALETTE[1],
s=70, alpha=0.8, edgecolors="white")
ax_res2.axhline(0, color="red", linestyle="--", lw=1.5)
ax_res2.set_title("SVR Residual Plot", fontweight="bold")
ax_res2.set_xlabel("Predicted Box Office (Cr)")
ax_res2.set_ylabel("Residual (Actual - Predicted)")

# Historical Box Office Trend
ax_trend = fig2.add_subplot(gs2[2, :])
df_sorted = df.sort_values("Year")
ax_trend.plot(df_sorted["Year"], df_sorted["BoxOffice_Cr"],
marker="o", color=PALETTE[0], lw=2, label="Box Office (Cr)",
zorder=3)
ax_trend.fill_between(df_sorted["Year"], df_sorted["BoxOffice_Cr"],
alpha=0.15, color=PALETTE[0])
for _, row in df_sorted.iterrows():
    if row["BoxOffice_Cr"] > 500:
        ax_trend.annotate(row["Title"], (row["Year"],
row["BoxOffice_Cr"]),
textcoords="offset points", xytext=(4, 6),
fontsize=7, color="darkblue")
ax_trend.set_title("Surya's Career – Box Office Timeline (1997-2025)",
fontweight="bold")
ax_trend.set_xlabel("Year")
ax_trend.set_ylabel("Box Office (Cr ₹)")

```

```

ax_trend.legend()

print("SECTION 9: PREDICTING KARUPPU BOX OFFICE")
print("-" * 60)

KARUPPU_BUDGET = 130.0
KARUPPU_RUNTIME = 155
KARUPPU_GENRES = "Action Drama"
KARUPPU_DIRECTOR = "RJ Balaji"
ASSUMED_RATING = 7.2
ASSUMED_VOTES = 18000

genre_flags = [1 if g in KARUPPU_GENRES else 0 for g in genre_categories]
director_bo_avg = df["BoxOffice_Cr"].mean()
karuppu_era = 5
years_since = 2026 - 1997

log_budget = np.log1p(KARUPPU_BUDGET)
log_votes = np.log1p(ASSUMED_VOTES)
budget_ratio = KARUPPU_BUDGET / (ASSUMED_RATING + 0.001)
rating_votes = ASSUMED_RATING * log_votes
num_genres = 2
is_multi = 0

karuppu_features = np.array([[
    ASSUMED_RATING, KARUPPU_RUNTIME, ASSUMED_VOTES,
    KARUPPU_BUDGET, director_bo_avg, karuppu_era, years_since,
    log_budget, log_votes, budget_ratio, rating_votes,
    num_genres, is_multi
] + genre_flags])

rf_pred_log = rf_pipeline.predict(karuppu_features)[0]
svr_pred_log = svr_pipeline.predict(karuppu_features)[0]

rf_pred_cr = np.expml(np.clip(rf_pred_log, 0, 20))
svr_pred_cr = np.expml(np.clip(svr_pred_log, 0, 20))
ensemble_cr = (rf_pred_cr * 0.6 + svr_pred_cr * 0.4)

rf_profit = rf_pred_cr - KARUPPU_BUDGET
svr_profit = svr_pred_cr - KARUPPU_BUDGET
ens_profit = ensemble_cr - KARUPPU_BUDGET

rf_roi = (rf_profit / KARUPPU_BUDGET) * 100
svr_roi = (svr_profit / KARUPPU_BUDGET) * 100
ens_roi = (ens_profit / KARUPPU_BUDGET) * 100

def verdict_from_roi(roi):
    if roi > 200: return "Blockbuster "
    elif roi > 100: return "Super Hit "
    elif roi > 50: return "Hit ,"
    elif roi > 0: return "Average "
    elif roi > -30: return "Flop "
    else: return "Disaster "

```

```

print(f"\n{'='*60}")
print(f"    KARUPPU – BOX OFFICE PREDICTION RESULTS")
print(f"    Director: {KARUPPU_DIRECTOR} | Budget: ₹{KARUPPU_BUDGET} Cr")
print(f"{'='*60}")
print(f"    Random Forest Prediction : ₹{rf_pred_cr:>8.2f} Cr | Profit:
₹{rf_profit:>7.2f} Cr | ROI: {rf_roi:>6.1f}% →
{verdict_from_roi(rf_roi)}")
print(f"    SVR Prediction          : ₹{svr_pred_cr:>8.2f} Cr | Profit:
₹{svr_profit:>7.2f} Cr | ROI: {svr_roi:>6.1f}% →
{verdict_from_roi(svr_roi)}")
print(f"    Ensemble (60RF+40SVR)      : ₹{ensemble_cr:>8.2f} Cr | Profit:
₹{ens_profit:>7.2f} Cr | ROI: {ens_roi:>6.1f}% →
{verdict_from_roi(ens_roi)}")
print(f"{'='*60}")

# Karuppu Prediction Chart
ax_pred = fig2.add_subplot(gs2[3, 0])
models = ["Random\nForest", "SVR", "Ensemble\n(Weighted)"]
pred_vals = [rf_pred_cr, svr_pred_cr, ensemble_cr]
bar_cols = [PALETTE[0], PALETTE[1], PALETTE[2]]
bars2 = ax_pred.bar(models, pred_vals, color=bar_cols, alpha=0.87,
                    edgecolor="white", width=0.5)
ax_pred.axhline(KARUPPU_BUDGET, color="red", linestyle="--",
                lw=2, label=f"Budget = ₹{KARUPPU_BUDGET} Cr")
for bar, val in zip(bars2, pred_vals):
    ax_pred.text(bar.get_x() + bar.get_width()/2,
                bar.get_height() + 5,
                f"₹{val:.1f} Cr", ha="center", fontsize=10,
                fontweight="bold")
ax_pred.set_title("Karuppu – Box Office Prediction\nby Model",
                fontweight="bold")
ax_pred.set_ylabel("Predicted Box Office (Cr ₹)")
ax_pred.legend(fontsize=9)

# Profit / ROI Chart
ax_roi = fig2.add_subplot(gs2[3, 1])
profits = [rf_profit, svr_profit, ens_profit]
roi_pct = [rf_roi, svr_roi, ens_roi]
bar_roi = ax_roi.bar(models, profits,
                    color=["#27ae60" if p > 0 else "#e74c3c" for p in
profits],
                    alpha=0.87, edgecolor="white", width=0.5)
ax_roi.axhline(0, color="black", lw=1.5)
for bar, roi_v, profit_v in zip(bar_roi, roi_pct, profits):
    ax_roi.text(bar.get_x() + bar.get_width()/2,
                max(bar.get_height(), 0) + 3,
                f"ROI: {roi_v:.1f}%\n₹{profit_v:.1f} Cr",
                ha="center", fontsize=9, fontweight="bold",
                color="#27ae60" if profit_v > 0 else "#e74c3c")
ax_roi.set_title("Karuppu – Profit & ROI Forecast\nby Model",
                fontweight="bold")
ax_roi.set_ylabel("Profit (Cr ₹)")

```



```

plt.savefig(SAVE_DIR + "page2_features_prediction.png", dpi=150,
bbox_inches="tight",
            facecolor="white")
plt.show()
print(", Page 2 saved: page2_features_prediction.png")

print("\n" + "=" * 60)
print("  FINAL SUMMARY TABLE")
print("=" * 60)
summary = pd.DataFrame({
    "Metric"      : ["R2 (Test)", "R2 (Train)", "CV R2 Mean", "MAE (Cr)", "RMSE (Cr)", "MAPE (%)"],
    "Random Forest" : [f"{rf_res['r2']:.4f}", f"{rf_res['train_r2']:.4f}", f"{rf_res['cv_mean']:.4f}", f"{rf_res['mae']:.2f}", f"{rf_res['rmse']:.2f}", f"{rf_res['mape']:.2f}"],
    "SVR"          : [f"{svr_res['r2']:.4f}", f"{svr_res['train_r2']:.4f}", f"{svr_res['cv_mean']:.4f}", f"{svr_res['mae']:.2f}", f"{svr_res['rmse']:.2f}", f"{svr_res['mape']:.2f}"],
})
print(summary.to_string(index=False))

print("\n" + "=" * 60)
print("  KARUPPU PREDICTION SUMMARY")
print("=" * 60)
karuppu_summary = pd.DataFrame({
    "Model"          : ["Random Forest", "SVR", "Ensemble (60/40)"],
    "Predicted BO (Cr)" : [f"₹{rf_pred_cr:.2f}", f"₹{svr_pred_cr:.2f}", f"₹{ensemble_cr:.2f}"],
    "Budget (Cr)"      : ["₹130", "₹130", "₹130"],
    "Net Profit (Cr)"   : [f"₹{rf_profit:.2f}", f"₹{svr_profit:.2f}", f"₹{ens_profit:.2f}"],
    "ROI (%)"          : [f"{rf_roi:.1f}%", f"{svr_roi:.1f}%", f"{ens_roi:.1f}%"],
    "Expected Verdict"  : [verdict_from_roi(rf_roi), verdict_from_roi(svr_roi), verdict_from_roi(ens_roi)]
})
print(karuppu_summary.to_string(index=False))

```

DATA SOURCES

1. IMDb. (n.d.). *Internet Movie Database*. <https://www.imdb.com>
2. Box Office Mojo. (n.d.). *Box office revenue and movie performance data*.
<https://www.boxofficemojo.com>
3. Kaggle. (n.d.). *Open-source datasets for data science and machine learning*.
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